

The Syndrome Weight Distribution in Quasi-Cyclic Codes, Applications to BIKE and HQC

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Abstract. Many important code-based cryptographic schemes such as the NIST post-quantum competition finalist BIKE and the to be standardized HQC scheme rely on Quasi-Cyclic Moderate-Density Parity-Check codes (QC-MDPC). A very important issue here is to predict accurately the Decoding Failure Rate (DFR). This DFR is intimately connected to the syndrome weight distribution of the QC-MDPC codes used in these schemes. This problem is treated in HQC by modeling the syndrome bits by Bernoulli variables which is known to be inaccurate. The rationale is that it gives a pessimistic estimate of the DFR. In BIKE the syndrome weight is modeled by the syndrome weight of a regular MDPC code which is itself computed by a simplified model. The accuracy of this modeling is not well understood. NIST perceived that BIKE DFR estimation lacked maturity. This led to its dismissal in the competition. The purpose of this paper is to advance on this difficult issue of understanding the syndrome weight distribution of quasi-cyclic codes. Our contribution here is threefold. First we provide a rigorous tool for computing the syndrome weight of a regular code through a generating function and a saddle point approximation. We use this approach to show that the Markov chain model used for estimating the syndrome weight in [ABP24a] is remarkably accurate. Second, we also prove that the regular model is not accurate for very low syndrome weights and provide a complete model of the syndrome weight distribution of a QC-MDPC code which can at the same time be computed quickly and fits remarkably well the experiments. We use this to show that for BIKE the probability of the events where the regular model differs from the QC-MDPC syndrome distribution is too low to be of concern. We also show that the variance of the syndrome weight distribution of a QC-MDPC code can be computed efficiently and is a handy tool for estimating accurately the syndrome weight distribution in the moderate deviation regime. We use it to give an accurate prediction of the DFR for a given key of HQC. This gives compelling evidence that the DFR of a typical secret key of HQC is significantly below $2^{-\lambda}$ where λ is the security parameter and that weak keys for HQC are too rare to be of concern.

1 Introduction

1.1 The Issue of Syndrome Weight in Code-based Cryptography

QC-MDPC Codes in Cryptography. Binary QC-MDPC codes are linear codes defined by a parity-check matrix, *i.e.* the code $\mathcal{C}$ is defined as $\mathcal{C} = \{\mathbf{x} \in \mathbb{F}_2^n : \mathbf{x}\mathbf{H}^\top = \mathbf{0}\}$ which is at the same time formed by blocks that are binary circulant matrices of a certain size and the number of 1's in each row and column is of order $O(\sqrt{n})$. The introduction of QC-MDPC codes in cryptography [MTSB13] building upon previous work [BBC08] gave a credible alternative to the McEliece cryptosystem [McE78, ABC⁺22] by basically providing much lower key sizes due to the quasi-cyclic structure. In the case of the McEliece cryptosystem which relies on a hidden algebraic structure due to the use of Goppa codes, this strategy of reducing the key size [BCGO09, MB09, BBB⁺17, BBB⁺17] initiated in [BCGO09] following the approach of [Gab05] by using quasi-cyclic Goppa codes gave rise to concerning algebraic attacks [FOPT10, BC18] that are partially explained by the fact that in this case the key recovery problem can be reduced to recover a much smaller Goppa code [FOP⁺16, Bar18]. In any case, the span of these attacks has to be better understood. However, in the case of HQC [AAB⁺22b], whose security relies solely on the decisional decoding problem of a generic quasi-cyclic code, or BIKE [AAB⁺22a], whose security relies solely on decoding a generic quasi-cyclic code and deciding whether or not a quasi-cyclic code has moderate weight codewords, moving to quasi-cyclic codes does not seem to weaken the scheme.

The Issue with the DFR. This strategy of using QC-MDPC codes has been followed for instance by the NIST post-quantum competition candidate **LEDACrypt** [BBC⁺19], the fourth round finalist **BIKE** [AAB⁺22a], and such codes also appear in **HQC** [AAB⁺22b] which is currently being standardized. However, using QC-MDPC codes in this setting has a price when compared to the McEliece cryptosystem, there is some nonzero probability that decryption fails due to a decoding failure. When IND-CCA2 security is required, the standard security proof [HHK17] requires the average DFR to be below $2^{-\lambda}$ where λ is the security parameter. The target DFR of $2^{-\lambda}$ is too small to be estimated experimentally and needs some theoretical proof. In the case of **HQC** the DFR can be directly linked to the weight of the syndrome $\mathbf{r}_1\mathbf{H}_1^\top + \mathbf{r}_2\mathbf{H}_2^\top$ of an error $(\mathbf{r}_1 \mathbf{r}_2)$ of weight $O(\sqrt{n})$ for the underlying QC-MDPC code with parity-check matrix $(\mathbf{H}_1 \mathbf{H}_2)$, where n is the code length. More precisely the error $\mathbf{e}'$ that the concatenated code used in **HQC** has to correct is of the form $\mathbf{e}' = \mathbf{r}_1\mathbf{H}_1^\top + \mathbf{r}_2\mathbf{H}_2^\top + \mathbf{e}$ where $\mathbf{e}$ is an error of weight $O(\sqrt{n})$ which is chosen independently of $(\mathbf{r}_1 \mathbf{r}_2)$. Decoding failure occurs when the weight of $\mathbf{e}'$ is too large and can not be corrected by the concatenated code. In the case of **BIKE** or **LEDACrypt**, decoding starts by computing the syndrome of an error of similar weight and even if the relationship between this syndrome weight and the DFR is less direct, it is essential in both cases to obtain an accurate prediction of the distribution of the syndrome weight to be able to predict the DFR.

In **HQC** [AAB⁺22b], the modeling of $\mathbf{e}'$ assumes that its bits are i.i.d random variables with Bernoulli distribution of parameter $p = \mathbf{P}(e'_i = 1)$. This modeling is in no way accurate as shown in [BRW25] but it is argued that this modeling leads to an underestimation of the actual DFR. In the case of **BIKE** [ALM⁺25], the modeling of the syndrome weight is done through a generating function approach taking into account indirectly the quasi-cyclic structure by incorporating information on near codewords which are specific to the quasi-cyclic structure. The computation of the relevant distribution is very expensive: for **BIKE** parameters of level 1, the computation takes more than 12 hours while using 52.5 GB of RAM on a 24 core server. The approach put forward in [ABP24b, ABPP25] does not take into account the quasi-cyclic structure at all by modeling $\mathbf{H}$ as random matrix of fixed row weight and column weight and arguing that the DFR obtained by this approach gives a pessimistic estimation of the DFR corresponding to a QC-MDPC code. See [ABP24b, §IV, p. 20] “Our estimate is however conservative, as the average DFR of the QC-LDPC codes was noted to be lower than their regular counterparts in [21]” or [ABPP25, §5, p.31] “The Figure highlights that our model, when employed for QC-MDPCs, provides a slightly conservative (i.e., higher) DFR estimate, as it is expected from literature results stating that random QC-MDPCs achieve slightly better correction capabilities w.r.t. their regular counterparts.”. This is highly questionable in the case of the QC-MDPC codes when it comes to estimate very small error probabilities. Here, as demonstrated in [ALM⁺25], there is an error floor behavior that is caused by the convergence of the decoder to near codewords that are very specific to the quasi-cyclic case. It is worthwhile to note that in the case of [ABP24b, ABPP25], the computation of the syndrome distribution is done through an efficient Markov chain approach for which there is experimental evidence that it provides an accurate approximation of the syndrome weight distribution for regular MDPC codes.

Worst-Case to Average-Case Reduction. The need to understand the syndrome distribution of QC-MDPC codes also arises from other issues in code-based cryptography. In particular a fundamental result in lattice-based cryptography, namely worst-case to average-case reduction between various lattice problems [Reg05, LPR10, RSW18], which provide very strong evidence for the hardness of the SIS or the SVP problems for instance, has been adapted to code-based cryptography in [BLVW19, YZ21, BCD23]. These articles provide a reduction from the worst-case search decoding problem to the decision version of the decoding which is often used in security proofs. The nice feature of [BCD23] is that it is a direct reduction from the first problem to the second one. This reduction holds in the unstructured case and the proof technique which is used is based on the adaptation to the code-based setting of a very general technique called OHCP [PRS17]. It was introduced [PRS17] in the lattice-based setting to also get reductions in the structured case (*i.e.* for Ring-LWE). However, to perform a similar reduction in the quasi-cyclic

code-based setting, we lack a good understanding of the distribution of syndromes for QC-MDPC codes (where MDPC has to be understood with the more general meaning of sufficiently sparse parity-check matrix), see [BCD23, §5].

1.2 Contributions

The QC-MDPC codes used in cryptography all have a similar regular structure, their parity-check matrix $\mathbf{H}$ is formed by a constant number of circulant blocks which all have the same weight. This structure induces what we call a (w, d) -regular code, namely a code with constant row weight w and constant column weight d . Understanding the syndrome weight distribution of regular codes is probably the simplest task to be performed before moving to the more complicated regular QC-MDPC code case. Surprisingly, even computing rigorously the syndrome weight distribution for typical regular MDPC codes is a non-trivial task and as far as we know does not appear in the literature. [SV19] and [ABP24a] introduces two different models which are experimentally accurate, but it would be desirable to obtain a rigorous computation. There are a few models of regular parity-check matrices that are much more amenable to a rigorous analysis such as the Gallager model [Gal63, §2.2, p.13] or the Standard LDPC model [RU08, §3.4, Def. 3.15 p.78]. However, they behave a little bit differently concerning the syndrome weight distribution when compared to the plain regular model mentioned above which is the closest to the quasi-cyclic model.

Rigorous Computation of the Syndrome Weight Distribution for regular codes. Our first contribution is a rigorous computation of the expected number of errors of a given weight t that have a syndrome of weight s for a code defined by a parity-check matrix uniformly distributed over the set of matrices of fixed row weight w and column weight d . This is obtained by a generating function approach based on a counting argument of matrices with a certain regular structure which is due to [BBK72]. The computation is really fast by using a saddle point approximation which bypasses the computation of the whole generating function. This gives a very efficient and rigorous method for computing the (average) syndrome distribution of a regular code. We use it to verify what is obtained in [SV19] and the Markov chain approach of [ABP24b, ABPP25] in cases which are out of reach of experimental studies and confirm that it gives a very accurate model in general.

Application to the QC-MDPC Setting. Obtaining the expected number of errors of a given weight t that have a syndrome of weight s for a random QC-MDPC code whose parity-check matrix is chosen uniformly at random with a constant number of blocks of some fixed weight seems out of reach of this kind of method. However, the quasi-cyclic structure can be partly taken into account by incorporating information on near codewords that appear to dominate the error floor behavior, as demonstrated in [ALM⁺25]. In this case, there are certain errors patterns of weight $\Theta(\sqrt{n})$ where n is the code length that have a very small syndrome weight, namely of order $\Theta(\sqrt{n})$. This is the case in BIKE whose parity-check matrix is formed by two circulant blocks of fixed row and column weight d . These patterns are just one column of one of those circulant blocks with 0's added in the remaining part. They are of weight d and their syndrome also turns out to be of weight d which is abnormally small. This structure generalizes to parity-check matrices which are formed by any number of circulant blocks. This concerns both HQC and LEDACrypt. In the case of the iterative decoding algorithm used in BIKE it appears that, in the error floor regime, decoding failures are well explained by the convergence to one of those near codewords [ALM⁺25].

From the mere fact that they have such a low syndrome weight, it is tempting to conjecture that the syndrome distribution is governed for small syndromes by error patterns that are too close to such a near codeword. Experiments confirm this intuition and this approach was also followed in [ALM⁺25] by conditioning the error by the event “the closest near codeword intersects the error in u positions”. We will follow the same approach here but improve on [ALM⁺25] by combining it with our generating function approach and saddle point approximation technique which makes the computation of the distribution of the syndrome weight conditioned on the value u much

more efficient and also more accurate. This allows to predict “a syndrome floor” behavior which is neither predicted by the aforementioned syndrome weight distribution of random regular codes that is obtained here, nor is it predicted by the Markov chain approach of [ABP24b, ABPP25] but which matches experimental evidence (see Fig. 3.4). This shows that the approach of approximating the syndrome weight distribution of QC-MDPC codes by the syndrome weight distribution of a random regular MDPC code of the relevant degrees is not accurate for very low syndrome weights. Fortunately, we also prove that for the **BIKE** level 1 parameter this “syndrome floor” behavior kicks in for syndrome weights that have a probability of occurring which is way below $2^{-\lambda}$ where λ is the security parameter. In other words, the probability of this event is too low to be of concern.

The Variance of the Syndrome Weight Distribution of QC-MDPC codes and Gaussian Approximation. Combining the regular approach with the lower bound with the syndrome distribution coming from conditioning the syndrome weight with respect to the size u of the intersection of the error with the closest near codeword allows a good prediction of the syndrome weight for a random QC-MDPC code. For **BIKE** or **LEDACrypt** the main issue is to predict correctly the distribution of syndromes below the average syndrome weight since low weight syndromes are the ones causing decoding failures. Taking into account the effect of near codewords is likely to be enough to predict this as shown in [ALM⁺25]. The situation is the opposite for **HQC**. Here the DFR is due to large syndrome weights and large syndrome weights are apparently not explained by the error being close to a certain pattern. We have found a strong correlation between unusually high probabilities for large syndrome weights and another quantity, which is the variance of the syndrome weight distribution associated to $(\mathbf{H}_1 \ \mathbf{H}_2)$. This variance can be rather easily derived from the parity-check matrix $(\mathbf{H}_1 \ \mathbf{H}_2)$ of the QC-MDPC code used in **HQC**. This is Proposition 4.6. It turns out that this variance being high is correlated with having an abnormally large probability for the syndrome weight to be large. This can be partly understood from the fact that the syndrome weight can be viewed as a sum of nearly independent random variables. This suggests that the syndrome weight could be approximated by a Gaussian distribution whose expectation and variance are respectively the expected value and variance of the syndrome weight. This approximation appears to be really accurate for moderate deviations. This explains why the probability of large syndrome weights is increasing with this variance.

The DFR of HQC. It is worthwhile to note that, by considering the variance, we have a powerful tool for assessing secret keys of **HQC**: keys $(\mathbf{H}_1 \ \mathbf{H}_2)$ that have an unusually large variance for the syndrome weight are also keys that display an unusually high DFR. We use Gaussian approximation for the syndrome weight to assess the DFR of a given secret key of **HQC**. It turns out that the DFR obtained by this approach is significantly below $2^{-\lambda}$ for a typical key. The estimation of the DFR for **HQC** was done by the aforementioned Bernoulli modeling and it was claimed in [AAB⁺22b] to give a conservative estimate of the DFR. We provide here an accurate estimate of this DFR by using Gaussian approximation based on computing the variance of the weight of $\mathbf{e}'$ for the noise term $\mathbf{e}' = \mathbf{r}_1 \mathbf{H}_1^\top + \mathbf{r}_2 \mathbf{H}_2^\top + \mathbf{e}$ that the concatenated code used in **HQC** has to correct. It shows that the DFR is about 2^{-147} for a typical **HQC** key. However, we have also observed that the variance over the key of the variance over the error of the syndrome weight is dramatically higher for a QC-MDPC code than for a regular MDPC code. This suggests that there might be weak keys in this setting which could have an abnormally large probability of having large weight syndromes. We first quantify this phenomenon experimentally by showing that the worst key we have found among $2.2 \cdot 10^{10}$ secret keys gives a DFR of $2^{-146.357}$. In other words, this study provides some strong evidence that the DFR of **HQC** was chosen very conservatively and that the scheme should be immune against attacks targeting high values of the DFR. It also points out that now the DFR of **HQC** is better understood, there is some room for improving the parameters of **HQC** a bit.

2 Background

2.1 Notation

Basic notation. Vectors and matrices are respectively denoted in bold letters and bold capital letters such as $\mathbf{a}$ and $\mathbf{A}$. Vectors are assumed to be row vectors and $\mathbf{x}^\top$ denotes the column vector which is the transpose of the row vector $\mathbf{x}$. The concatenation of two matrices $\mathbf{A}$ and $\mathbf{B}$ with the same number of rows side by side is denoted by $(\mathbf{A} \ \mathbf{B})$. For a matrix $\mathbf{A}$ and a subset I of column indices of $\mathbf{A}$, we denote by $\mathbf{A}_I$ the submatrix of $\mathbf{A}$ formed by the columns of $\mathbf{A}$ whose indices belong to I . The entry at index i of the vector $\mathbf{x}$ is denoted by x_i . Sometimes we also write $\mathbf{x}(i)$ for it such as in $\mathbf{eH}(i)$, meaning the i -th entry of the vector $\mathbf{eH}$. The Hamming weight $|\mathbf{x}|$ of a vector $\mathbf{x}$ is its number of nonzero entries. If x is a real number and $\mathbf{v} = (v_i)_{0 \leq i \leq n-1}$ a vector of $\mathbb{R}^n$, $x + \mathbf{v}$ and $x - \mathbf{v}$ stand for the vectors $(x + v_i)_{0 \leq i \leq n-1}$ and $(x - v_i)_{0 \leq i \leq n-1}$ respectively. For such a binary vector $\mathbf{e} = (e_i)_{0 \leq i \leq n-1}$ we denote by $\bar{\mathbf{e}} = (1 - e_i)_{0 \leq i \leq n-1}$, *i.e.* the binary vector with complementary support. We may also write for a position i and a binary vector $\mathbf{e}$ that $i \in \mathbf{e}$ if and only if i is in the support of $\mathbf{e}$. We will often identify a binary vector with its support and for instance for a set of positions I and a binary vector $\mathbf{e}$ write $I \cap \mathbf{e}$ to mean the intersection of the set of positions with the support of $\mathbf{e} = (e_0, \dots, e_{n-1})$ which is $\{i \in \llbracket 0, n-1 \rrbracket : e_i = 1\}$. For binary vectors $\mathbf{x}$ and $\mathbf{y}$, we also denote by $\mathbf{x} \cap \mathbf{y}$ the intersection of the support of $\mathbf{x}$ with the support of $\mathbf{y}$. For a matrix $\mathbf{A} \in \mathbb{F}_2^{r \times n}$ and a vector $\mathbf{x} \in \mathbb{F}_2^n$ we denote by $\mathbf{A}_{\mathbf{x}}$ the submatrix of $\mathbf{A}$ formed by the columns of $\mathbf{A}$ whose indices belong to the support of $\mathbf{x}$. For a polynomial $p(x) = \sum_{i=0}^d p_i x^i$ we denote by $\text{deg sup}(p)$ its degree support, that is $\text{deg sup}(p) = \{i \in \mathbb{N} : p_i \neq 0\}$. For two integers a and b such that $a \leq b$, $\llbracket a, b \rrbracket$ denotes the set of integers in the interval $[a, b]$, namely $\{a, a+1, \dots, b\}$.

Binary linear codes. All codes considered here are binary and linear, which means that they are vector spaces over $\mathbb{F}_2$. A code $\mathcal{C}$ of length n and dimension k is a k -dimensional subspace of $\mathbb{F}_2^n$. We say that such a code is an $[n, k]$ code. It is specified by a parity-check matrix which is a matrix whose right kernel is $\mathcal{C}$, *i.e.* $\mathcal{C} = \{\mathbf{c} \in \mathbb{F}_2^n : \mathbf{cH}^\top = \mathbf{0}\}$.

Quasi-cyclic codes and polynomial notation. On top of being binary and linear, certain codes considered here are quasi-cyclic, *i.e.* their parity-check matrix is formed by blocks of circulant matrices. We will only be interested in matrices that are of the form $\mathbf{H} = (\mathbf{C}_0 \ \mathbf{C}_1 \ \dots \ \mathbf{C}_{\text{idx}-1})$ where all the $\mathbf{C}_i$'s are circulant matrices. idx is referred to as the index of the quasi-cyclic code. Such codes are conveniently represented by polynomials since $r \times r$ binary circulant matrices form a ring $\mathcal{R}$ isomorphic to $\mathbb{F}_2[x]/(x^r - 1)$, the isomorphism Ψ being given by

$$\begin{pmatrix} c_0 & c_{r-1} & \dots & c_2 & c_1 \\ c_1 & c_0 & c_{r-1} & & c_2 \\ \vdots & c_1 & c_0 & \ddots & \vdots \\ c_{r-2} & \vdots & \ddots & \ddots & c_{r-1} \\ c_{r-1} & c_{r-2} & \dots & c_1 & c_0 \end{pmatrix} \mapsto c_0 + c_1 x + \dots + c_{r-1} x^{r-1}.$$

We use this isomorphism to view the matrix $\mathbf{H} = (\mathbf{C}_0 \ \dots \ \mathbf{C}_{\text{idx}-1})$ as the idx -tuple $(h_0(x), \dots, h_{\text{idx}-1}(x))$ where $h_i(x) = \Psi(\mathbf{C}_i)$. We likewise represent error vectors $\mathbf{e} = (e_0, \dots, e_{n-1}) \in \mathbb{F}_2^n$ where $n = \text{idx} \cdot r$ with the following idx -tuple of polynomials

$$\mathbf{e} \rightsquigarrow (e_0(x), \dots, e_{\text{idx}-1}(x)) \in (\mathbb{F}_2[x]/(x^r - 1))^{\text{idx}}$$

where $e_i(x) = e_{ir} + \dots + e_{i(r-1)} x^{r-1}$. The syndrome $\mathbf{s} = (s_0, \dots, s_{r-1})$ of an error $\mathbf{e} = (e_0, \dots, e_{n-1})$, *i.e.* $\mathbf{s} = \mathbf{eH}^\top$ is also represented as a polynomial $s(x) = \sum_{j=0}^{r-1} s_j x^j$ and

$$s(x) = e_0(x)h_0(x) + \dots + e_{a-1}(x)h_{a-1}(x). \quad (2.1)$$

Note that all polynomial operations are performed in the ring $\mathbb{F}_2[x]/(x^r - 1)$.

Notation 1. From now on we will always assume that the quasi-cyclic code we are interested in has a parity-check matrix $\mathbf{H}$ given in polynomial form by

$$\mathbf{h} = (h_0(x), \dots, h_{\text{idx}-1}(x))$$

where idx always stands for the index of the quasi-cyclic code.

It will be helpful to stick to the case which is considered in HQC and BIKE, namely that we assume that the size r of the circulant blocks is always an odd prime. This choice avoids for instance attacks coming from the fact that $1 + x^a$ is a divisor of $1 + x^{a \cdot b}$.

Assumption 1. The block size r of the circulant blocks in all the quasi-cyclic codes considered in this paper is assumed to be an odd prime.

2.2 Near codewords

Near codewords were introduced in the literature of iterative decoding as a way to formalize a collection of error vectors with nonzero lower-than-expected syndrome weight and that are therefore difficult to decode.

Definition 2.1 (near codeword of type (s, t)). An error vector $\mathbf{e} \in \mathbb{F}_2^n$ is a near codeword of type (s, t) if $|\mathbf{e}| = t$ and $|\mathbf{e}\mathbf{H}^\top| = s$.

MDPC codes are known to have minimum distance which is typically linear in the code length n and are unlikely to have small near codewords. This is not the case for QC-MDPC codes: the minimum distance is of order $O(\sqrt{n})$. This leads one to suspect that such codes also have small-weight near codewords. This is indeed the case and this is most likely the main issue for decoding as identified in [Vas21]:

Notation 2. Let $\mathbf{h} = (h_0(x), \dots, h_{\text{idx}-1}(x)) \in (\mathbb{F}_2[x]/(x^r - 1))^{\text{idx}}$. We define

$$\nu_{\ell,b}(\mathbf{h}) \stackrel{\text{def}}{=} (0, \dots, 0, \underbrace{x^\ell h_b(x)}_{b\text{-th position}}, 0, \dots, 0).$$

Definition 2.2 (The set $\mathcal{N}$ of near codewords). Consider a quasi-cyclic code of index idx with circulant blocks of size r with parity-check matrix in polynomial form $\mathbf{h} = (h_0(x), \dots, h_{\text{idx}-1}(x))$. The set $\mathcal{N}$ of near codewords is given in polynomial form by

$$\mathcal{N} \stackrel{\text{def}}{=} \bigsqcup_{0 \leq b < \text{idx}} \mathcal{N}_b, \text{ where } \mathcal{N}_b \stackrel{\text{def}}{=} \{\nu_{\ell,b}(\mathbf{h}), \ell \in \llbracket 0, r-1 \rrbracket\},$$

Proposition 2.3. Consider a quasi-cyclic code with circulant blocks that are all of constant column (and row) weight d . The elements of $\mathcal{N}$ are near codewords of type (d, d) .

Proof. We suppose that the quasi-cyclic code is given in polynomial form by $\mathbf{h} = (h_0(x), \dots, h_{\text{idx}-1}(x))$. This proposition follows from the fact that the syndrome of $\nu_{\ell,b}(\mathbf{h})$ is equal to $x^\ell h_b(x)^2$ which is readily seen to have exactly d nonzero coefficients since in $\mathbb{F}_2[x]/(x^r - 1)$ squaring and multiplication by x preserve the number of nonzero coefficients. The fact that squaring preserves the Hamming weight is a direct consequence of the fact that r is odd. $\square$

In this work, when we refer to a near codeword, we mean an element of $\mathcal{N}$.

Unusually low syndrome weight. Elements of $\mathcal{N}$ are (d, d) -near codewords as per Def. 2.1: they are of weight d and have syndrome weight d . Moreover, even errors $\mathbf{e}$ which have an overlap of size u with a near codeword ν of this kind (meaning that $|\mathbf{e} \cap \nu| = u$) have an unusually low syndrome weight. We namely have :

Proposition 2.4 (Low syndrome weight). *Consider a quasi-cyclic code of index idx with circulant blocks of size r that are all of constant column (and row) weight d . Let ν be an element of $\mathcal{N}$ and let $\mathbf{e}$ be an error of weight t . Let $u \stackrel{\text{def}}{=} |\mathbf{e} \cap \nu|$. The syndrome of $\mathbf{e}$ is of weight $\leq d \cdot t - u(u-1)$.*

Proof. Let $\mathbf{h} = (h_0, \dots, h_{\text{idx}-1}(x))$ be the polynomial form of the parity-check matrix of the quasi-cyclic code. Without loss of generality we may assume that the near codeword ν is say $\nu_{\ell,0}(\mathbf{h}) = (x^\ell h_0(x), 0, \dots, 0)$ for $\ell \in \llbracket 0, r-1 \rrbracket$. Let $(e_0(x), e_1(x), \dots, e_{\text{idx}-1}(x))$ be the polynomial form of the error $\mathbf{e}$. We may write $e_0(x)$ and $x^\ell h_0(x)$ as $e_0(x) = a(x) + b(x)$, $x^\ell h_0(x) = a(x) + c(x)$ where $a(x)$ is the polynomial for the common part $\mathbf{e} \cap \nu$, i.e. $a(x) = \sum_{i: e_i = h_{(i-\ell) \bmod r}} x^i$. By using (2.1), we know that the syndrome $s(x)$ of $\mathbf{e}$ is

$$\begin{aligned} s(x) &= e_0(x)h_0(x) + \sum_{i=1}^{\text{idx}-1} e_i(x)h_i(x) \\ &= x^{r-\ell}(a(x) + b(x))(a(x) + c(x)) + \sum_{i=1}^{\text{idx}-1} e_i(x)h_i(x) \\ &= x^{r-\ell}(a(x)^2 + a(x)b(x) + a(x)c(x) + b(x)c(x)) + \sum_{i=1}^{\text{idx}-1} e_i(x)h_i(x). \end{aligned} \quad (2.2)$$

Recall here that we perform the polynomial computation over $\mathbb{F}_2[x]/(x^r - 1)$ and that $x^{r-\ell}x^\ell = 1$ in this ring. Let us denote by $|P|$ the weight of a polynomial P , namely the number of its nonzero coefficients. The weight satisfies $|P \cdot Q| \leq |P| \cdot |Q|$ and $|P + Q| \leq |P| + |Q|$ for all polynomials P and Q . Notice that $d \cdot t = d \cdot \sum_{i=0}^{\text{idx}-1} |e_i|$ is exactly $|e_0| \cdot |h_0| + \sum_{i=1}^{\text{idx}-1} |e_i| \cdot |h_i|$, which in turn is exactly $|a|^2 + |a| \cdot |b| + |a| \cdot |c| + |b| \cdot |c| + \sum_{i=1}^{\text{idx}-1} |e_i| \cdot |h_i|$ because $|e_0| = |a| + |b|$ and $|h_0| = |x^\ell h_0| = |a| + |c|$. The point is that $|a^2| = |a|$ because squaring preserves the number of nonzero coefficients. This remark together with (2.2) yields $|s| \leq |a^2| + |a| \cdot |b| + |a| \cdot |c| + |b| \cdot |c| + \sum_{i=1}^{\text{idx}-1} |e_i| \cdot |h_i| = |a| + |a| \cdot |b| + |a| \cdot |c| + |b| \cdot |c| + \sum_{i=1}^{\text{idx}-1} |e_i| \cdot |h_i|$, we have at least a drop of $|a|^2 - |a| = u(u-1)$ in the syndrome weight as claimed above. $\square$

2.3 Tanner graphs

It will be convenient to use here the language of *Tanner graphs*.

Definition. A Tanner graph is a very handy tool for analyzing iterative decoding of LDPC or MDPC codes. For LDPC codes, this notion dates back to Gallager who explained and studied his iterative decoding algorithms [Gal63] by using them. In a more general form, they have been defined in [Tan81]. It is a bipartite graph which represents the parity-check matrix $\mathbf{H} = (H_{i,j})_{\substack{0 \leq i < r \\ 0 \leq j < n}}$ of a code. It has two types of vertices, the *variable nodes* which are in bijection with the code positions $\{0, \dots, n-1\}$, and the *check nodes* which are in bijection with the parity checks, i.e. the rows of $\mathbf{H}$. There is an edge between a variable node j and the check node i if and only if $H_{i,j} = 1$.

Tanner graph of a quasi-cyclic code. In the case of a quasi-cyclic code of index idx with parity-check matrix $(h_0(x), \dots, h_{\text{idx}-1}(x))$ we may index a variable node by the pair (i, b) where $i \in \llbracket 0, r-1 \rrbracket$ and $b \in \llbracket 0, \text{idx}-1 \rrbracket$ and it is readily verified that the check nodes adjacent to (i, b) are all the check nodes labeled $(i+j) \bmod r$ where j is any element in the degree support $\text{deg}_{\text{sup}}(h_b)$ of h_b . Particular subgraphs of this Tanner graph will play a prominent role, these are the subgraphs of the Tanner graph induced by a near codeword. From now on we will use the following notation.

Notation 3. The subgraph of the Tanner graph induced by a near codeword ν , in other words the subgraph of the Tanner graph formed by the variable nodes belonging to the support of ν and all the check nodes and the edges of the Tanner graph linking these vertices, is denoted by $\mathcal{G}(\nu)$. The sequence of degrees in $\mathcal{G}(\nu)$ of the check nodes is denoted by $\Delta(\nu)$, this is the sequence $(\Delta_i)_{0 \leq i < r}$ where Δ_i is the degree of the check node i in $\mathcal{G}(\nu)$.

The structure of these subgraphs is given by

Proposition 2.5. Consider a quasi-cyclic code of index $\text{id}\mathbf{x}$ and block length r with parity-check in polynomial form $\mathbf{h} = (h_0(x), \dots, h_{\text{id}\mathbf{x}-1}(x))$. The subgraph of its Tanner graph induced by the near codeword in polynomial form $\nu_{\ell,b}(\mathbf{h}) = (0, \dots, 0, \underbrace{x^\ell h_b(x)}_{b\text{-th position}}, 0, \dots, 0)$ has variables nodes (i, b)

where $i \in \{(j + \ell) \bmod r : j \in \text{deg}\text{sup}(h_b)\}$ and there is an edge from (i, b) to any $(i + j) \bmod r$ such that j is in $\text{deg}\text{sup}(h_b)$.

3 Efficient Computation of the Average Number of Errors with a Given Syndrome Weight

In this section, we provide an efficient method for computing the average number of errors of a given weight t that have a syndrome of a given weight s . The average is taken over all regular codes of some given row and column weights. Our approach is inspired from the proof of [LS02, Theorem 4]. We provide in a first step a generating function whose coefficients are these averages. Then we provide an efficient method for computing these coefficients based on a saddle point approximation. This is then tweaked to estimate the same quantity for quasi-cyclic codes by taking partially into account their structure. In this case, we observe experimentally for low syndrome weight values a “syndrome floor” behavior that our method predicts. We apply this to BIKE. It speeds up very significantly the first step of the DFR estimation computation in [ALM⁺25], which was previously the bottleneck of the computation. We also show that this syndrome floor is too low for BIKE to affect the DFR estimates.

3.1 Asymptotic Counting of Matrices with Given Weight Sequences

In this subsection, we introduce the main tool for computing the average syndrome weight distribution, it consists in theorems and corollaries counting binary matrices with prescribed weight sequences (or, equivalently, simple bipartite graphs with prescribed degree sequences).

Notation 4. Let n and r be two positive integers and let $\mathbf{w} = (w_1, \dots, w_r) \in \llbracket 0, n \rrbracket^r$ and $\mathbf{d} = (d_1, \dots, d_n) \in \llbracket 0, r \rrbracket^n$ be two integer vectors. We define

- $\Lambda_{r,n}^{\mathbf{w},\mathbf{d}}$ to be the set of matrices in $\mathbb{F}_2^{r \times n}$ with row weights $w_1, \dots, w_r$ and column weights $d_1, \dots, d_n$ respectively.
- In this notation, when we just have scalars instead of vectors, such as $\Lambda_{r,n}^{w,d}$, this denotes the set of (w, d) -regular matrices, meaning that they have constant row weight w and constant column weight d .

We will use the following asymptotic equivalent from [BBK72, Theorem 2] :

Theorem 3.1. Let w, d be fixed integers. For $r, n \in \mathbb{N}$ and $\mathbf{w} \in \llbracket 0, w \rrbracket^r, \mathbf{d} \in \llbracket 0, d \rrbracket^n$ such that $\sum_{i=1}^r w_i = \sum_{j=1}^n d_j$,

$$|\Lambda_{r,n}^{\mathbf{w},\mathbf{d}}| \underset{E \rightarrow +\infty}{\sim} \frac{E!}{\prod_{i=1}^r w_i! \prod_{j=1}^n d_j!} \exp \left(- \frac{[\sum_{i=1}^r w_i(w_i - 1)] \cdot [\sum_{j=1}^n d_j(d_j - 1)]}{2E^2} \right)$$

where E denotes the sum $\sum_{i=1}^r w_i = \sum_{j=1}^n d_j$.

Remark 3.2. E can be viewed as the number of edges of the Tanner graph associated to a matrix in $\Lambda_{r,n}^{\mathbf{w},\mathbf{d}}$ (which is itself viewed as a parity-check matrix).

More specifically, we need the following corollaries :

Corollary 3.3. *Let w, d be fixed integers. For $r, n, t \in \mathbb{N}$ such that $rw = nd$, and $\mathbf{j} \in \llbracket 0, w \rrbracket^r$ such that $\sum_{i=1}^r j_i = dt$,*

$$\frac{|\Lambda_{r,t}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-t}^{w-\mathbf{j},d}|}{|\Lambda_{r,n}^{w,d}|} \underset{n,t \rightarrow +\infty}{\sim} \frac{\prod_{i=1}^r \binom{w}{j_i} \exp\left(-\left[\frac{j_i(j_i-1)}{dt} + \frac{(w-j_i)(w-j_i-1)}{d(n-t)}\right] \frac{d-1}{2}\right)}{\binom{dn}{dt} \exp\left(-\frac{(w-1)(d-1)}{2}\right)}$$

Corollary 3.4. *Let w, d be fixed integers. For $r, n, t \in \mathbb{N}$, and $\mathbf{w}, \mathbf{j} \in \llbracket 0, w \rrbracket^r$ such that $0 \leq j_i \leq w_i$ for $1 \leq i \leq r$, $\sum_{i=1}^r w_i = dn$ and $\sum_{i=1}^r j_i = dt$,*

$$\frac{|\Lambda_{r,t}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-t}^{\mathbf{w}-\mathbf{j},d}|}{|\Lambda_{r,n}^{\mathbf{w},d}|} \underset{n,t \rightarrow +\infty}{\sim} \frac{\prod_{i=1}^r \binom{w_i}{j_i} \exp\left(-\left[\frac{j_i(j_i-1)}{dt} + \frac{(w_i-j_i)(w_i-j_i-1)}{d(n-t)}\right] \frac{d-1}{2}\right)}{\binom{dn}{dt} \exp\left(-\frac{\sum_{i=1}^r w_i(w_i-1)}{dn} \frac{d-1}{2}\right)}$$

We will also use the notation below in what follows

Notation 5. $a_j^{(n,w,d,t)} \stackrel{\text{def}}{=} \binom{w}{j} \exp\left(-\left[\frac{j(j-1)}{dt} + \frac{(w-j)(w-j-1)}{d(n-t)}\right] \frac{d-1}{2}\right).$

Notation 6.

$$f^{(n,w,d,t)}(X) \stackrel{\text{def}}{=} \sum_{0 \leq j \leq w} a_j^{(n,w,d,t)} X^j$$

Notation 7. *For a polynomial $f(X_1, \dots, X_m) = \sum_{j_1, \dots, j_m} f_{j_1, \dots, j_m} X_1^{j_1} \dots X_m^{j_m}$,*

$$f_{\text{even}}(X_1, \dots, X_m) \stackrel{\text{def}}{=} \sum_{j_1 + \dots + j_m \in 2\mathbb{N}} f_{j_1, \dots, j_m} X_1^{j_1} \dots X_m^{j_m}$$

$$f_{\text{odd}}(X_1, \dots, X_m) \stackrel{\text{def}}{=} \sum_{j_1 + \dots + j_m \notin 2\mathbb{N}} f_{j_1, \dots, j_m} X_1^{j_1} \dots X_m^{j_m}.$$

Notation 8. *For a polynomial $f \in \mathbb{R}[X_1, \dots, X_m]$ and a tuple $(k_1, \dots, k_m) \in \mathbb{N}^m$, $[X_1^{k_1} \dots X_m^{k_m}]f(X_1, \dots, X_m)$ denotes the coefficient associated with the monomial $X_1^{k_1} \dots X_m^{k_m}$.*

3.2 The Distribution of Syndrome Weights of Random Regular Codes

Recall that a regular code of type (w, d) is a code which has a parity-check matrix belonging to $\Lambda_{r,n}^{w,d}$ where n is the codelength and r the number of rows of this parity-check matrix. The distribution of the syndrome weight of such a regular code is given by

Theorem 3.5. *Let w, d be fixed integers. Let $r, n, t \in \mathbb{N}$ be such that $rw = nd$. Let $\mathbf{H}$ and $\mathbf{e}$ be random variables uniformly distributed over $\Lambda_{r,n}^{w,d}$ and the set of binary vectors of length n and Hamming weight t respectively. Let $S = |\mathbf{eH}^\top|$. If $s - dt \equiv 1 \pmod{2}$, then $\mathbf{P}(S = s) = 0$ and otherwise we have*

$$\mathbf{P}(S = s) \underset{n,t \rightarrow +\infty}{\sim} \frac{b_{dt,s}^{(n,w,d,t)}}{\binom{dn}{dt} \exp\left(-\frac{(w-1)(d-1)}{2}\right)}$$

where

$$\begin{aligned} b_{j,s}^{(n,w,d,t)} &\stackrel{\text{def}}{=} [X^j Y^s] \left(\left[f_{\text{even}}^{(n,w,d,t)}(X) + Y \cdot f_{\text{odd}}^{(n,w,d,t)}(X) \right]^r \right) \\ &= [X^j] \left(\binom{r}{s} \left[f_{\text{odd}}^{(n,w,d,t)}(X) \right]^s \left[f_{\text{even}}^{(n,w,d,t)}(X) \right]^{r-s} \right). \end{aligned}$$

3.3 Quasi-Cyclic Codes

We can use a regular model, in our case the asymptotic equivalent given in §3.1, to estimate the syndrome distribution of a quasi-cyclic code as it is done for instance in [ABP24b, ABPP25]. However, it turns out that for small values of the syndrome weight we cannot ignore specific behaviors coming from the quasi-cyclicity and the existence of near codewords specific to this case. Indeed, for a regular quasi-cyclic code of index idx , block length r and constant block weight d , there is a set $\mathcal{N} = \bigsqcup_{0 \leq b < \text{idx}} \mathcal{N}_b$ of $\text{idx} \cdot r$ (d, d) -near codewords which have been identified in [Vas21] and that are recalled in Definition 2.2 and in Proposition 2.3.

The important fact is that the Tanner graph of the code restricted to the positions of a near codeword $\nu \in \mathcal{N}$ has a very particular structure as explained in Proposition 2.5. By using this proposition we readily deduce that

Proposition 3.6. *For a given quasi-cyclic code of index idx and $b \in \llbracket 0, \text{idx} - 1 \rrbracket$, the degree distribution of the subgraph induced by the support of a near codeword ν is the same for all ν 's belonging to the same $\mathcal{N}_b$.*

In other words, this degree distribution depends only on the block b . Many of the results we are going to present in what follows depend only on this degree distribution. We will give a notation for it.

Notation 9. *The degree distribution $\mathbf{n}(b) = (n_i)_{0 \leq i \leq d}$ of block b is defined by*

$$n_i = |\{j \in \llbracket 0, r - 1 \rrbracket \mid \Delta_j = i\}|$$

where $\Delta = \Delta(\nu_{0,b})$ is the degree sequence of the near codeword $\nu_{0,b}$ corresponding to block b .

To simplify the discussion about the Tanner graph structure we are going to study, we fix a block b of the quasi-cyclic code and index the variable nodes belonging to this block just by $0, \dots, r - 1$ instead of $(0, b), (1, b), \dots, (r - 1, b)$. We also consider the near codeword $\nu = (0, \dots, 0, \underbrace{h_b(x)}_{b\text{-th position}}, 0, \dots, 0)$. Its support is therefore the set of positions $\text{deg}\text{sup}(h_b)$. The sub-

graph of the Tanner graph induced by the variables nodes in $\text{deg}\text{sup}(h_b)$ is given by the edges linking a variable node $i \in \text{deg}\text{sup}(h_b)$ to a check node labelled $i + j \bmod r$ where j is in $\text{deg}\text{sup}(h_b)$. In the simplest case where all the $i + j \bmod r$ for $i, j \in \text{deg}\text{sup}(h_b)$ with $i \leq j$ are distinct, there are d nodes of degree 1 (corresponding to $k = i + i \bmod r$), $\frac{d(d-1)}{2}$ check nodes k of degree 2 (corresponding to $k = i + j \bmod r$, $i \neq j$) and the remaining $r - \frac{d(d+1)}{2}$ nodes have degree 0. We call *perfect keys* the codes with this degree distribution for all blocks $b \in \{0, \dots, \text{idx} - 1\}$. Typical keys are not perfect but still have very constrained degree distributions.

Given an error pattern $\mathbf{e}$, the pairs $(i, j) \in \nu \times \nu$, $i \leq j$ are divided into five subsets :

$$\begin{aligned} - P_1 &\stackrel{\text{def}}{=} \{(i, j) \mid i, j \in \nu, i < j, |\{i, j\} \cap \mathbf{e}| = 2\} & - P_4 &\stackrel{\text{def}}{=} \{(i, i) \mid i \in \nu \cap \mathbf{e}\} \\ - P_2 &\stackrel{\text{def}}{=} \{(i, j) \mid i, j \in \nu, i < j, |\{i, j\} \cap \mathbf{e}| = 1\} & - P_5 &\stackrel{\text{def}}{=} \{(i, i) \mid i \in \nu \cap \bar{\mathbf{e}}\} \\ - P_3 &\stackrel{\text{def}}{=} \{(i, j) \mid i, j \in \nu, i < j, |\{i, j\} \cap \mathbf{e}| = 0\} \end{aligned}$$

Lemma 3.7 (Cardinalities). *Let $u = |\nu \cap \mathbf{e}|$.*

$$\begin{aligned} - |P_1| &= \binom{u}{2} & - |P_3| &= \binom{d-u}{2} & - |P_5| &= d - u \\ - |P_2| &= u(d - u) & - |P_4| &= u \end{aligned}$$

For a perfect key, these sets are identified with subsets of the parity-check equations, of degree 2 for P_1, P_2, P_3 and degree 1 for P_4, P_5 . In this case, we also denote by P_6 the subset of parity-check equations with degree 0.

We will use the near codewords as follows to tweak the syndrome weight distribution of the regular codes to take into account the quasi-cyclicity. For this purpose, we will use the following

probabilistic model to approximate in the quasi-cyclic case $\mathbf{P}(S = s \mid U = u)$ with $\mathbf{P}(S' = s \mid U' = u)$ where $S = |\mathbf{e}\mathbf{H}^\top|$, $\mathbf{H}$ is the parity-check matrix of the quasi-cyclic code we are interested in, $U = |\nu \cap \mathbf{e}|$ and where $S' = |\mathbf{e}\mathbf{H}'^\top|$, $U' = |\nu' \cap \mathbf{e}|$ with $\mathbf{H}'$ being a random matrix drawn according to a certain probabilistic model, ν' is a fixed vector of the same weight as ν . More precisely we define our probabilistic model as

Model 1. Let $\Delta \in \llbracket 0, d \rrbracket^r$. Let $\mathbf{H}' = (\mathbf{H}'_0 \mathbf{H}'_1)$ where $\mathbf{H}'_0 \in \mathbb{F}_2^{r \times d}$ and $\mathbf{H}'_1 \in \mathbb{F}_2^{r \times (n-d)}$ are drawn uniformly at random in $\Lambda_{r,d}^{\Delta,d}$ and $\Lambda_{r,n-d}^{w-\Delta,d}$, respectively. Let $\nu' \in \mathbb{F}_2^n$ whose support is in the first d positions. We let $S' \stackrel{\text{def}}{=} |\mathbf{e}\mathbf{H}'^\top|$ and $U' \stackrel{\text{def}}{=} |\nu' \cap \mathbf{e}|$ where $\mathbf{e}$ is uniformly distributed over the binary words of length n and weight t .

We will make the following assumption

Assumption 2. We assume that for all pairs of integers s and u we have

$$\mathbf{P}(S = s \mid U = u) = \mathbf{P}(S' = s \mid U' = u)$$

where $S = \mathbf{e}\mathbf{H}^\top$, $U = |\nu \cap \mathbf{e}|$. Here $\mathbf{e}$ is uniformly distributed over the set of binary vectors of length n and weight t , whereas $\mathbf{H}$ is uniformly distributed over the set of matrices formed by concatenating side by side idx circulant matrices of size r and weight d that correspond to a quasi-cyclic code having a near codeword ν whose degree sequence $\Delta(\nu)$ satisfies $\Delta(\nu) = \Delta$.

The rationale behind this assumption, is that the quasi-cyclic distribution of S conditioned on the value of U is essentially governed by the degree sequence of the closest near codeword and if we just take into account this degree sequence in a corresponding regular model, we basically have a good model for the quasi-cyclic distribution. This philosophy will be backed up with compelling experimental evidence for large values of U as we show in §3.5.

Theorem 3.8. Let $\text{idx}, r, d, t, u, s$ be integers such that $0 \leq u \leq \min(d, t)$ and $0 \leq s \leq dt - u(u-1)$, $w = \text{idx} \cdot d$. Let $\Delta \in \llbracket 0, d \rrbracket^r$. Let $\mathbf{H}$ be a random variable uniformly distributed over the set of matrices formed by concatenating side by side idx circulant matrices of size r and weight d that correspond to quasi-cyclic code that have a near codeword ν whose degree sequence $\Delta(\nu)$ satisfies $\Delta(\nu) = \Delta$. Let $\mathbf{e}$ be a random variable uniformly distributed the set of binary vectors of length n and weight t . Let $U = |\nu \cap \mathbf{e}|$ and $S = |\mathbf{e}\mathbf{H}^\top|$. If $s - dt \equiv 1 \pmod{2}$, then $\mathbf{P}(S = s \mid U = u) = 0$ and otherwise under Assumption 2 we have

$$\mathbf{P}(S = s \mid U = u) = \sum_{\substack{0 \leq \mathbf{k} \leq \Delta \\ 0 \leq \mathbf{j} \leq w - \Delta \\ k_1 + \dots + k_r = du \\ j_1 + \dots + j_r = d(t-u) \\ |(\mathbf{k} + \mathbf{j}) \cap (2\mathbb{N} + 1)| = s}} \frac{|\Lambda_{r,u}^{\mathbf{k},d}| \cdot |\Lambda_{r,d-u}^{\Delta - \mathbf{k},d}|}{|\Lambda_{r,d}^{\Delta,d}|} \cdot \frac{|\Lambda_{r,t-u}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-d-t+u}^{w-\Delta-\mathbf{j},d}|}{|\Lambda_{r,n-d}^{w-\Delta,d}|}$$

We cannot rigorously prove the approximation under the same parameter setting as Theorem 3.5 because the weight of the submatrix induced by the positions of the near codeword, equal to d^2 , would be constant, so Corollary 3.4 for the asymptotic equivalent does not apply. Nevertheless, the approximation is accurate in practice for the parameter sets of interest. The approximation is as follows:

$$\begin{aligned}
\mathbf{P}(S = s \mid U = u) &= \sum_{\substack{0 \leq \mathbf{k} \leq \Delta \\ 0 \leq \mathbf{j} \leq w - \Delta \\ k_1 + \dots + k_r = du \\ j_1 + \dots + j_r = d(t-u) \\ |(\mathbf{k} + \mathbf{j}) \cap (2\mathbb{N} + 1)| = s}} \frac{|\Lambda_{r,u}^{\mathbf{k},d}| \cdot |\Lambda_{r,d-u}^{\Delta - \mathbf{k},d}|}{|\Lambda_{r,d}^{\Delta,d}|} \cdot \frac{|\Lambda_{r,t-u}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-d-t+u}^{w-\Delta-\mathbf{j},d}|}{|\Lambda_{r,n-d}^{w-\Delta,d}|} \\
&\approx \sum_{\substack{0 \leq \mathbf{k} \leq \Delta \\ 0 \leq \mathbf{j} \leq w - \Delta \\ k_1 + \dots + k_r = du \\ j_1 + \dots + j_r = d(t-u) \\ |(\mathbf{k} + \mathbf{j}) \cap (2\mathbb{N} + 1)| = s}} \frac{\prod_{i=1}^r a_{k_i}^{(d, \Delta_i, d, u)} a_{j_i}^{(n-d, w-\Delta_i, d, t-u)}}{\beta^{(n, w, d, t, u, \Delta)}} \\
&= \frac{b_{du, d(t-u), s}^{(n, w, d, t, u, \Delta)}}{\beta^{(n, w, d, t, u, \Delta)}}
\end{aligned}$$

where $b_{k,j,s}^{(n, w, d, t, u, \Delta)}$ is defined as :

$$[X^k Y^j Z^s] \left(\prod_{\Delta} \left[g_{\text{even}}^{(n, w, d, t, u, \Delta)}(X, Y) + Z \cdot g_{\text{odd}}^{(n, w, d, t, u, \Delta)}(X, Y) \right]^{n_{\Delta}} \right)$$

with $g^{(n, w, d, t, u, \Delta)}(X, Y) \stackrel{\text{def}}{=} f^{(d, \Delta, d, u)}(X) \cdot f^{(n-d, w-\Delta, d, t-u)}(Y)$
and $\beta^{(n, w, d, t, u, \Delta)}$ is defined as :

$$\left(\frac{d^2}{du} \right) \left(\frac{d(n-d)}{d(t-u)} \right) \exp \left(- \left[\frac{\sum_{i=1}^r \Delta_i (\Delta_i - 1)}{d^2} + \frac{\sum_{i=1}^r (w - \Delta_i)(w - \Delta_i - 1)}{d(n-d)} \right] \frac{d-1}{2} \right)$$

3.4 Fast Computation with Saddle Point Approximation

Our approximation framework stems from [RU08, Appendix D.4], in which we find the following results regarding the asymptotic behavior of coefficients in polynomial powers, coming from Cauchy's integral formula.

First, for univariate polynomials : [RU08, pp. 507-510]

Theorem 3.9 (Hayman's approximation for powers of univariate polynomials). *Let $P \in \mathbb{R}[X]$ be a non-constant polynomial with nonnegative coefficients and nonzero constant coefficient, of degree d , and $\alpha \in (0, d)$. For $n \in \mathbb{N}$ such that $k := \alpha n \in \mathbb{N}$, and $F := P^n$, we have :*

$$[X^k]F(X) \underset{n \rightarrow +\infty}{\sim} \frac{\delta \cdot F(x)}{x^k \sqrt{2\pi B(x)}}$$

where $A = X \frac{F'}{F}$, $B = X A'$, $x > 0$ is such that $A(x) = k$ and $\delta = \gcd\{k \in \mathbb{N} \mid [X^k]P(X) > 0\}$.

Such a polynomial is said to be *Hayman admissible* when $\delta = 1$.

Remark 3.10. If P is even and all the coefficients of even indexes are nonzero, then $\delta = 2$.

The authors generalize the notion of Hayman admissibility for multivariate polynomials [RU08, Def. D.5 and Lem. D.8] and prove that : [RU08, Lem. D.14]

Theorem 3.11 (Hayman's approximation for powers of multivariate polynomials). *Let $P \in \mathbb{R}[X_1, \dots, X_m]$ be a Hayman admissible polynomial, of degree d_i in X_i for $0 \leq i \leq m$, and $\alpha \in (0, d_1) \times \dots \times (0, d_m)$. For $n \in \mathbb{N}$ such that $\mathbf{k} := \alpha n \in \mathbb{N}^m$, and $F := P^n$, we have :*

$$[X_1^{k_1} \dots X_m^{k_m}]F(X_1, \dots, X_m) \underset{n \rightarrow +\infty}{\sim} \frac{F(x_1, \dots, x_m)}{x_1^{k_1} \dots x_m^{k_m} \sqrt{(2\pi)^m \det(\mathbf{B}(x_1, \dots, x_m))}}$$

where $A_i = X_i \frac{\partial_i F}{F}$ for $1 \leq i \leq m$, $B_{i,j} = X_j \partial_j A_i$ for $1 \leq i, j \leq m$, and $x_1, \dots, x_m > 0$ are such that $\mathbf{A}(x_1, \dots, x_m) = \mathbf{k}$.

In the case of our generating functions, we would need such a result in a slightly more general setting. If F is one of these functions, we assume that the following approximation holds, which is backed by experimental verification :

$$[X_1^{k_1}, \dots, X_m^{k_m}] F(X_1, \dots, X_m) \approx \frac{\delta \cdot F(x_1, \dots, x_m)}{x_1^{k_1} \dots x_m^{k_m} \sqrt{(2\pi)^m \det(\mathbf{B}(x_1, \dots, x_m))}}$$

where $A_i = X_i \frac{\partial_i F}{F}$ for $1 \leq i \leq m$, $B_{i,j} = X_j \partial_j A_i$ for $1 \leq i, j \leq m$, $x_1, \dots, x_m > 0$ are such that $\mathbf{A}(x_1, \dots, x_m) = \mathbf{k}$, and δ is some constant as in Theorem 3.9, whose value does not matter since we approximate probability distributions, and therefore always normalize the results.

3.5 Results

Regular codes. Figure 3.1 shows theoretical and experimental results for syndrome weight distribution for a regular square matrix of size $n \times n$ with small parameters $n = r = 101$, $w = d = 7$ and $t = 7$. We have chosen such a small example in order to be able to observe experimentally the probability of low syndrome weights. Note that in the case of such small parameters, we did not compute our theoretical formula using the saddlepoint method described in 3.4, but instead directly extracted the relevant coefficients from the generating function using SageMath. Even for such small parameters, the asymptotic theoretical model agrees really well with the syndrome weight distribution of a random regular matrix of constant row and column weight 7. Figure 3.2 compares [SV19] and [ABP24a] models with our regular model, as well as experiments in the region where they are available, for BIKE. There is no noticeable difference between (i) the approximate model [SV19], (ii) the approximate Markov model of [ABP24a], (iii) our asymptotic regular model. Moreover, all three models correspond to the experimental evidence in the region where we could compute the syndrome weight. In this case, we use a random quasi-cyclic code to perform the experiments.

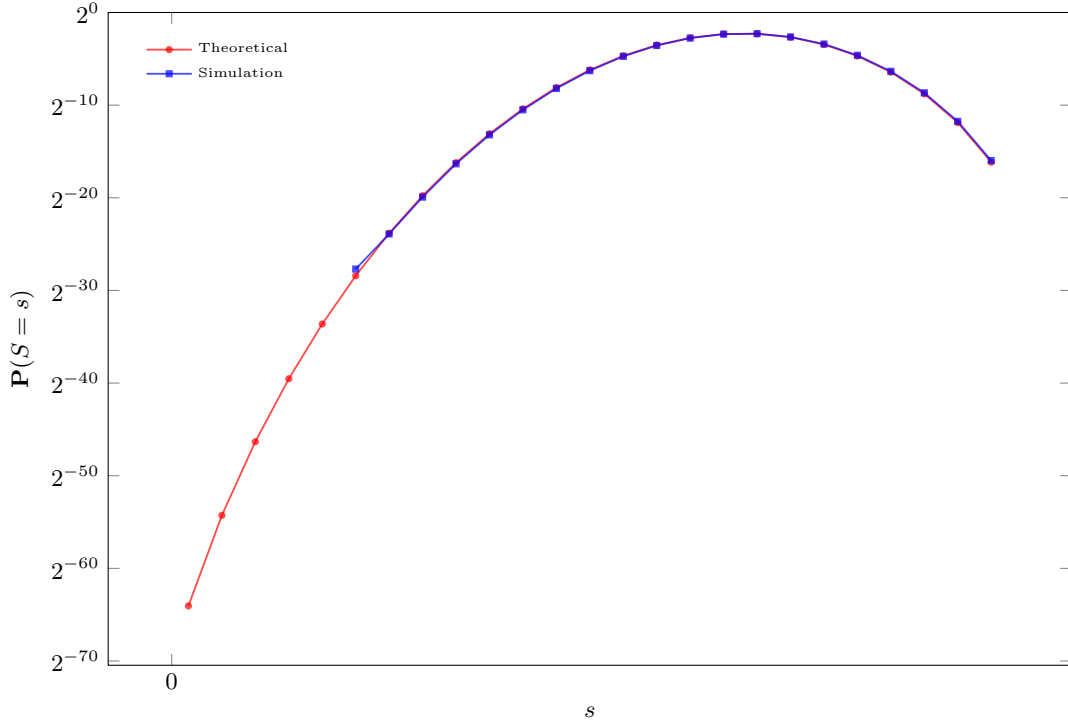


Fig. 3.1: Syndrome weight distribution for a regular block with $n = r = 101$, $w = d = 7$ and $t = 7$.

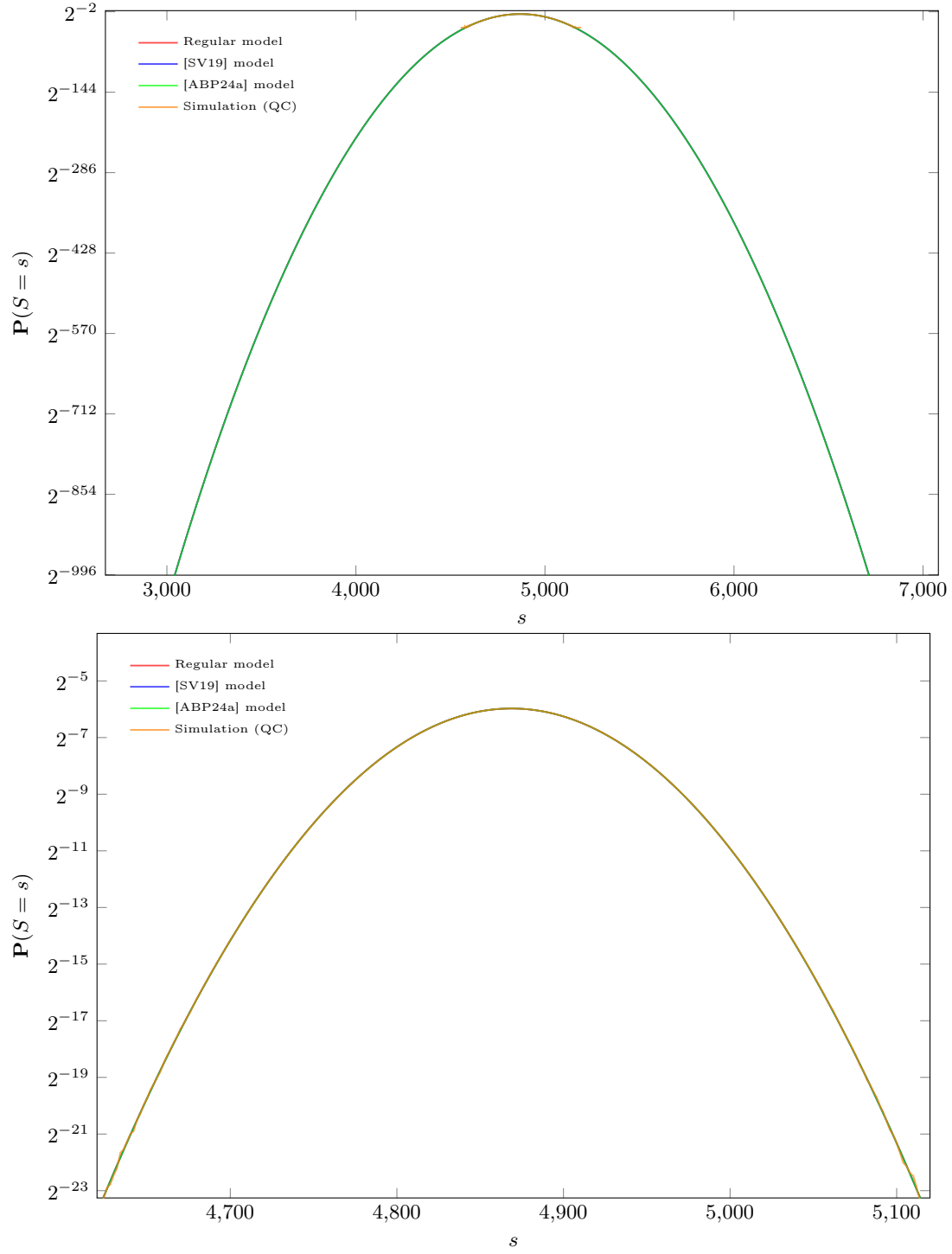


Fig. 3.2: Syndrome weight distribution for $(\text{idx}, r, d, t) = (2, 12323, 71, 134)$ (BIKE parameters)

Quasi-Cyclic Codes. For moderate deviations around the mean, experiments show that the syndrome weight distribution of a quasi-cyclic code is well approximated by the model of [SV19], the Markovian model used to approximate the syndrome distribution of an MDPC code [ABP24b] and the computation of the syndrome weight of a random regular code proposed here. However, we will show here that for small values of the syndrome weight, there is definitely a “syndrome floor” behavior which is not taken into account by these three models and which could be problematic for the estimation of the DFR. This is clearly demonstrated by using the results of Subsection 3.3 to derive a quasi-cyclic model which departs from the regular model for small syndrome values as shown in Figure 3.3. We have chosen here a random key for **BIKE** and have computed the estimation we have for the quasi-cyclic distribution. It shows a deviation from the regular model for syndrome weights below 3800. The corresponding probabilities are below 2^{-128} . This shows that we can still take the regular model for the syndrome weight model for **BIKE** but this is not something we could have decided without performing our study here. We have verified the quasi-cyclic model on small examples and have verified that we indeed predict with our quasi-cyclic model the experimental distribution of the syndrome weight. This is shown in Figure 3.4.

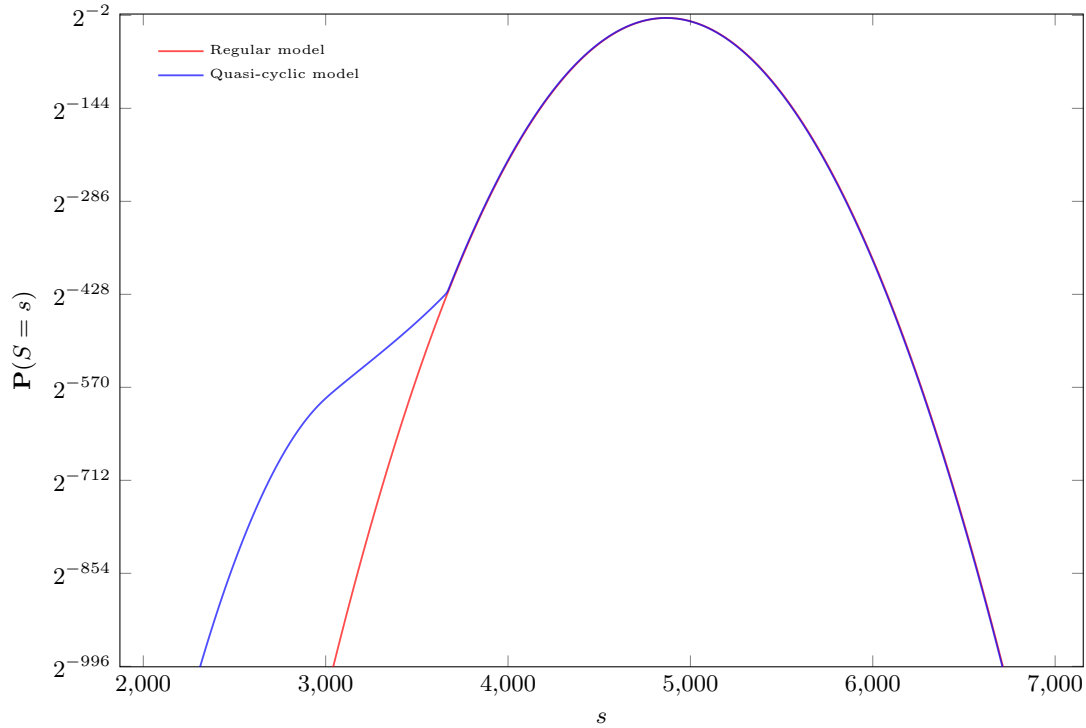


Fig. 3.3: Syndrome weight distribution for $(\text{idx}, r, d, t) = (2, 12323, 71, 134)$ and degree distribution $n_0 = 10000, n_1 = 59, n_2 = 2043, n_3 = 11, n_4 = 198, n_5 = 1, n_6 = 11$ for both blocks

Let us now explain how we obtained these results by using the results from §3.3. It turns out that when the error $\mathbf{e}$ has a large intersection with one of the near codewords of the quasi-cyclic code, then the syndrome weight will be abnormally small. To quantify for the cumulative effect of all near codewords, we bring in $U \stackrel{\text{def}}{=} \max\{|\nu \cap \mathbf{e}| \mid \nu \in \mathcal{N}\}$, the largest cardinality of intersection between a near codeword and the error, instead of the intersection with a particular near codeword. We compute the syndrome weight distribution using the law of total probability :

$$\mathbf{P}(S = s) = \sum_u \mathbf{P}(S = s \mid U = u) \mathbf{P}(U = u). \quad (3.1)$$

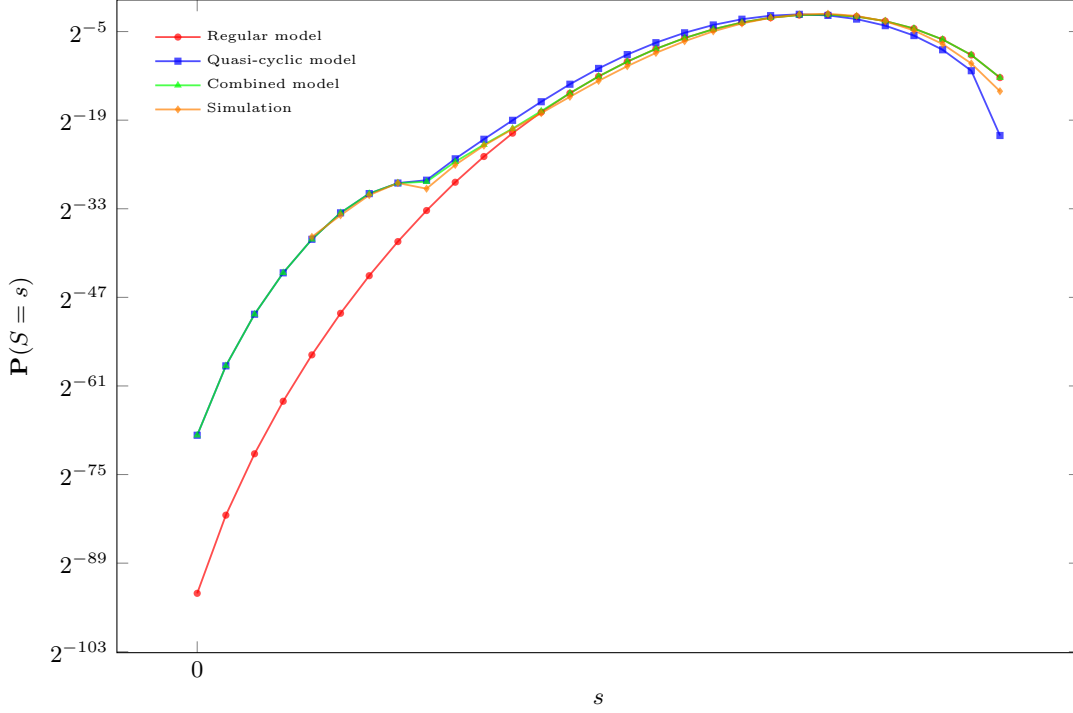


Fig. 3.4: Syndrome weight distribution for $(\text{idx}, r, d, t) = (1, 163, 7, 8)$, perfect keys.

In the quasi-cyclic case it turns out that when s is small enough, the sum (3.1) is dominated by the terms corresponding to a large value of u . This is what is causing the syndrome floor behavior that we will now quantify.

We derive the distribution of U by assuming that the random variables $|\nu \cap \mathbf{e}|$ for $\nu \in \mathcal{N}$ are independent. Note that the cardinality of $\mathcal{N}$ is nothing but the code length n . Let ρ_u , respectively $\tilde{\rho}_u$, be the probability that $|\nu \cap \mathbf{e}| = u$, respectively $|\nu \cap \mathbf{e}| \geq u$, for a particular near codeword $\nu \in \mathcal{N}$. Then we have that

$$\rho_u = \frac{\binom{d}{u} \binom{n-d}{t-u}}{\binom{n}{t}}, \quad \tilde{\rho}_u = \sum_{u' \geq u} \rho_{u'}.$$

From the independence assumption we deduce that

$$\mathbf{P}(U = u) = \mathbf{P}(U \geq u) - \mathbf{P}(U \geq u + 1) \text{ with } \mathbf{P}(U \geq u) = 1 - (1 - \tilde{\rho}_u)^n$$

or equivalently

$$\mathbf{P}(U = u) = \sum_{k=1}^n \binom{n}{k} (\rho_u)^k (1 - \tilde{\rho}_u)^{n-k}.$$

Figure 3.6 shows the results for BIKE-1 parameters on one block. We obtain an experimental lower bound using the following method : we can easily sample error patterns that have a large number of common positions u with a near codeword, and therefore experimentally measure the conditional distribution $\mathbf{P}(S = s \mid U = u)$ for these large u . Using the aforementioned model for $\mathbf{P}(U = u)$, this allows to obtain some terms of the sum (3.1) and thus a lower bound on $\mathbf{P}(S = s)$. The interest of this approach is that it allows to observe indirectly the “syndrome floor” region where large values of u dominate the sum (3.1).

The conditional distribution $\mathbf{P}(S = s \mid U = u)$ is well approximated by our model for large u , which means that $|\nu \cap \mathbf{e}|$ is atypical for only one near codeword (the one which has a large intersection with the error). When u is small, $|\nu \cap \mathbf{e}|$ is atypical for all near codewords, and here we

underestimate the syndrome weight. This means that this method correctly predicts the impact of near codewords on low syndrome weights, but is less accurate than the regular model in the typical region.

But the anomaly we want to predict (the leftmost part of the curve in Figure 3.6a) is induced by large values of u . Figure 3.7a shows $\mathbf{P}(U = u \mid S = s)$ for $s = 3806$, which is around the point where the regular and quasi-cyclic curves diverge. We observe that there are two main contributions, one from values of u around 6 and another from values of u around 51. The first one is well predicted by the regular model, and our quasi-cyclic model is useful to predict the second one. Thus, we shall define a threshold u_{th} , decompose $\mathbf{P}(S = s)$ as

$$\mathbf{P}(S = s \mid U \leq u_{\text{th}}) \mathbf{P}(U \leq u_{\text{th}}) + \sum_{u > u_{\text{th}}} \mathbf{P}(S = s \mid U = u) \mathbf{P}(U = u)$$

then compute $\mathbf{P}(S = s \mid U \leq u_{\text{th}})$ according to the regular model, and $\mathbf{P}(S = s \mid U = u)$ for $u > u_{\text{th}}$ according to the quasi-cyclic model.

In order to define the value of this threshold, we consider the function $s \mapsto \mathbb{E}(U \mid S = s)$, represented on Figure 3.7b. It is essentially flat for low and high syndrome weights, and it sharply decreases around the point where the regular and quasi-cyclic curves diverge. We define u_{th} as the ordinate of the inflection point, which yields $u_{\text{th}} \approx 28.84$ in our example.

We call this approach the “combined model”. Figure 3.4 shows that it predicts remarkably well the syndrome floor behavior.

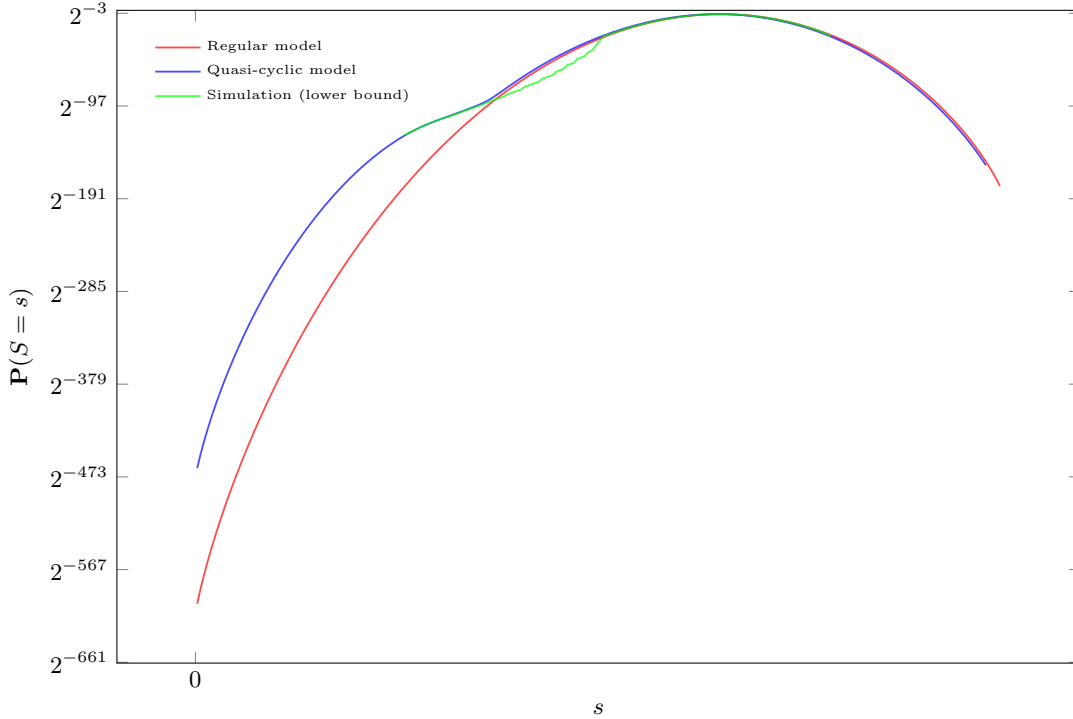
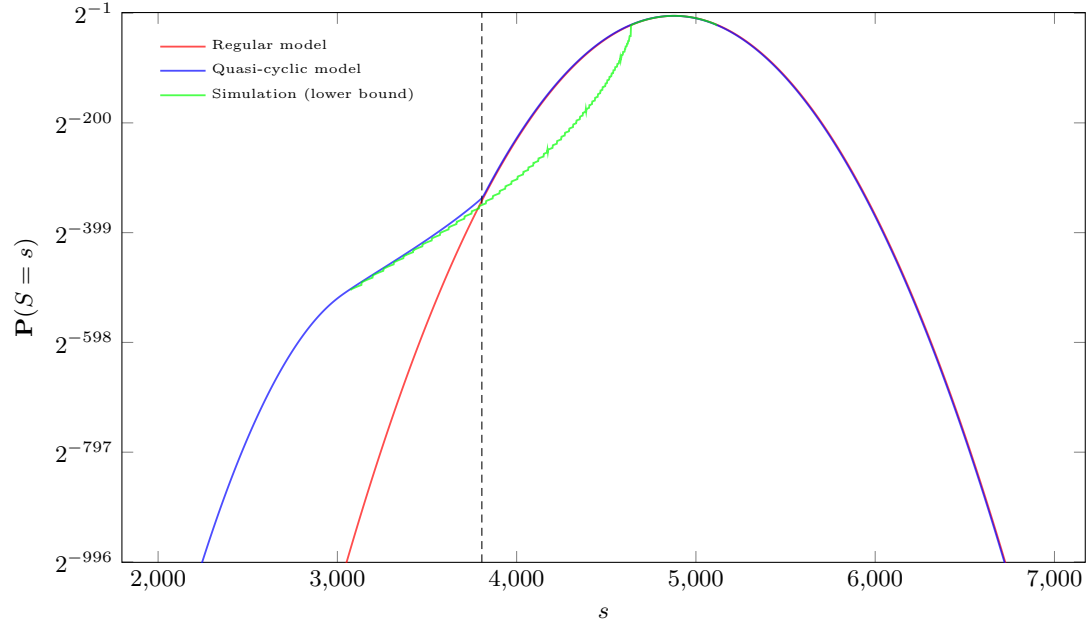


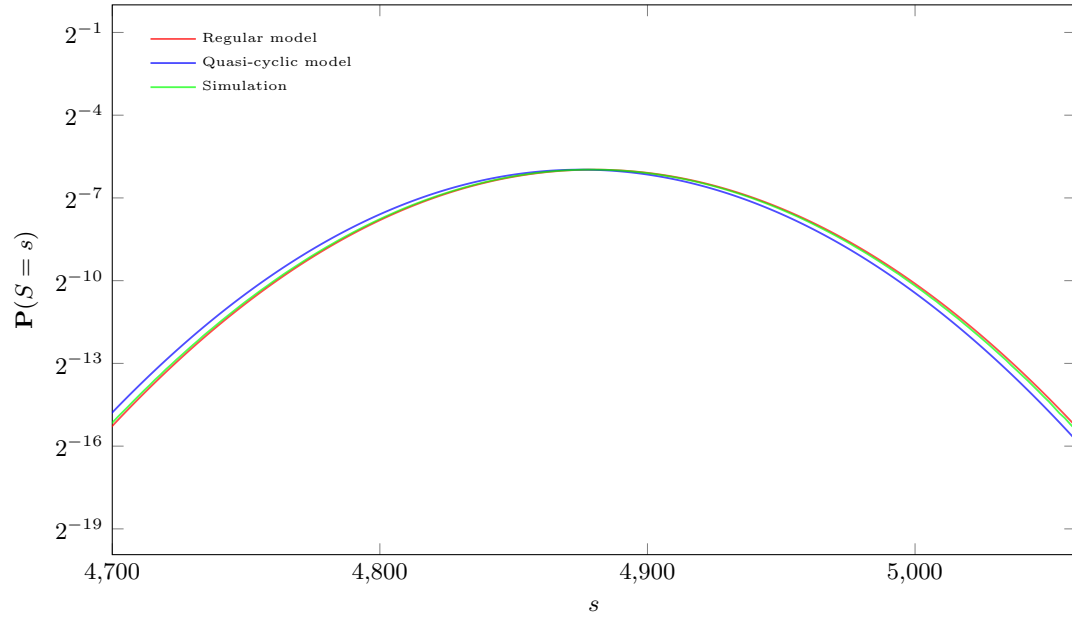
Fig. 3.5: Syndrome weight distribution for $(\text{idx}, r, d, t) = (2, 941, 17, 27)$

4 Prediction for Large Syndrome Weights

The analysis of the previous section showed that we have a good prediction of the syndrome distribution for values of syndrome below the mean by combining the regular model with the

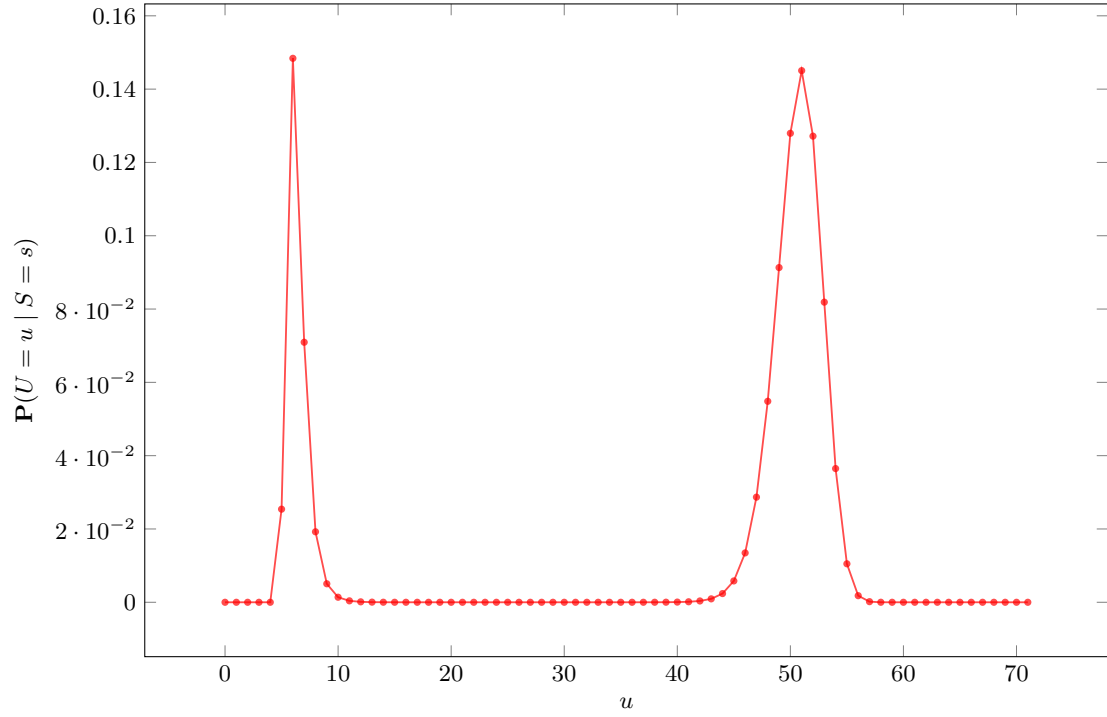


(a)

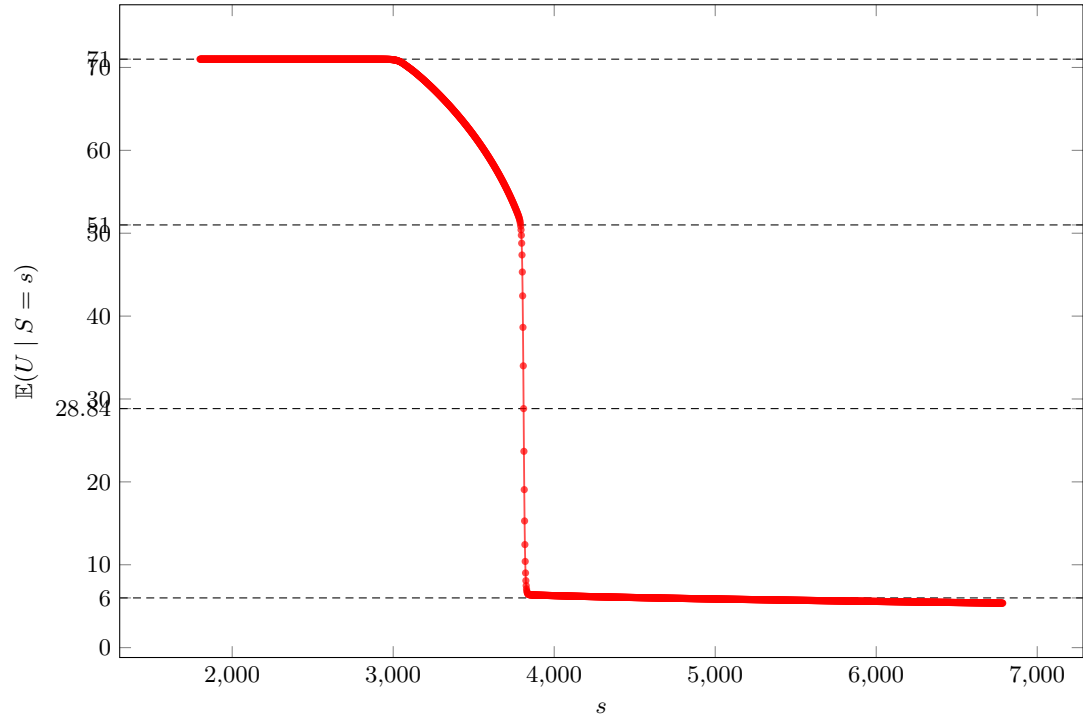


(b)

Fig. 3.6: Syndrome weight distribution for $(\text{idx}, r, d, t) = (1, 12323, 71, 134)$ and degree distribution $n_0 = 10000$, $n_1 = 59$, $n_2 = 2043$, $n_3 = 11$, $n_4 = 198$, $n_5 = 1$, $n_6 = 11$



(a) $\mathbf{P}(U = u \mid S = s)$ for $s = 3806$ according to our model for $(\mathbf{id}\mathbf{x}, r, d, t) = (1, 12323, 71, 134)$ and degree distribution $n_0 = 10000, n_1 = 59, n_2 = 2043, n_3 = 11, n_4 = 198, n_5 = 1, n_6 = 11$



(b) $\mathbb{E}(U \mid S = s)$ according to our model for $(\mathbf{id}\mathbf{x}, r, d, t) = (1, 12323, 71, 134)$ and degree distribution $n_0 = 10000, n_1 = 59, n_2 = 2043, n_3 = 11, n_4 = 198, n_5 = 1, n_6 = 11$

quasi-cyclic model taking into account U , the maximum intersection size of the error with a near codeword. This model seems adequate for computing the DFR for BIKE. However, for HQC it is important to be able to assess the probability of having an abnormally large syndrome weight. Indeed, here the syndrome weight is the weight of the error that the decapsulation algorithm has to decode, therefore the failure probability is high for large weights. In the NIST PQC submission [AAB⁺22b] a simple binary symmetric channel model is used to estimate the DFR. It is argued that this model overestimates the probability of having large values of the syndrome weight and that this leads to an overestimation of the DFR for HQC. Our aim in this section is to have a much better model for the syndrome weight in the case of HQC and derive a better estimate of the DFR from it. By doing so, we will also be able to understand the proportion of weak keys which display a higher DFR. Our approach is to find a simple value which can be used to assess the distribution of the syndrome weight above the mean which is key dependent. It consists in considering the variance (over the set of errors) of the syndrome weight. From the mere fact that the normal approximation of the syndrome weight based on this variance turns out to be accurate for moderate deviations, it is clear that this variance should be a good predictor of a weak key. This is indeed confirmed by experimental evidence (see Figure 4.1) showing that the higher the variance, the larger the probability of having a large syndrome weight.

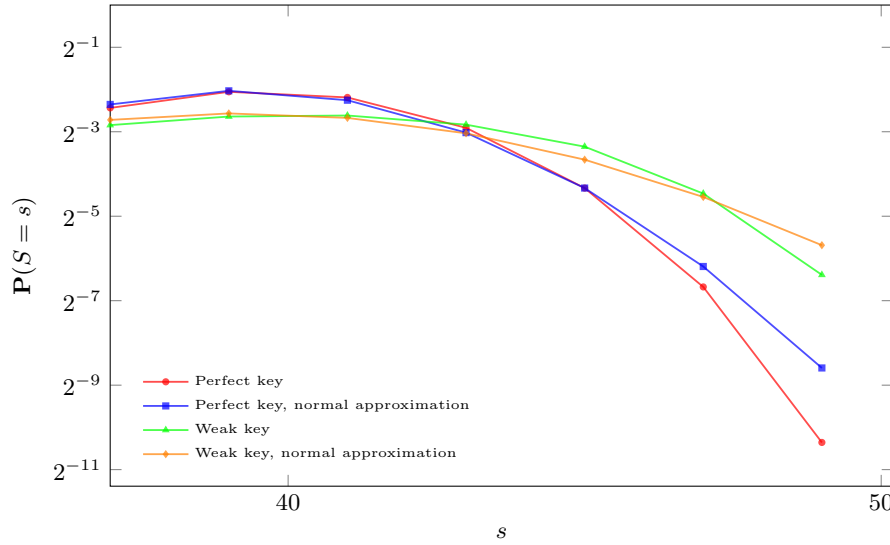


Fig.4.1: Experimental syndrome weight distributions and their normal approximation, for $(\text{id}\mathbf{x}, r, d, t) = (1, 163, 7, 7)$

4.1 Computation of the Variance of the Syndrome Weight

The Case of BIKE. In BIKE we are interested in understanding the weight S_{BIKE} where $\mathbf{H}$ is chosen as a random quasi-cyclic code of index $\text{id}\mathbf{x} = 2$ with two circulant blocks of column and row weight d and $\mathbf{e}$ is a random error of weight t

$$S_{\text{BIKE}} = |\mathbf{e}\mathbf{H}^\top|. \quad (4.1)$$

To compute the variance $\text{Var}[S_{\text{BIKE}}]$ (over $\mathbf{e}$) of S_{BIKE} we use the following lemmas

Lemma 4.1. Let $\mathbf{h} \in \mathbb{F}_2^n$ of Hamming weight w and assume that $\mathbf{e}$ is an error of Hamming weight t in $\mathbb{F}_2^n$ chosen uniformly at random. Let $p_n(t, w) \stackrel{\text{def}}{=} \mathbf{P}(\langle \mathbf{h}, \mathbf{e} \rangle = 1)$. We have

$$p_n(t, w) = \sum_{\substack{i=1 \\ i \text{ odd}}}^n \frac{\binom{w}{i} \binom{n-w}{t-i}}{\binom{n}{t}}.$$

Lemma 4.2. Let $\mathbf{h}$ and $\mathbf{h}'$ be two vectors of $\mathbb{F}_2^n$ of weight w . Let $a \stackrel{\text{def}}{=} |\mathbf{h} \cap \mathbf{h}'|$. Let $\mathbf{e}$ be an element of $\mathbb{F}_2^n$ of weight t drawn uniformly at random. We have

$$\text{Cov}(\langle \mathbf{h}, \mathbf{e} \rangle, \langle \mathbf{h}', \mathbf{e} \rangle) = \sum_{\substack{p, q, r, s \in \mathbb{N} \\ p+q \text{ odd} \\ p+r \text{ odd} \\ p+q+r+s=t}} \frac{\binom{a}{p} \binom{w-a}{q} \binom{w-a}{r} \binom{n-2w+a}{s}}{\binom{n}{t}} - p_n(t, w)^2.$$

We denote by $\Sigma_n(a, t, w)$ this expression for the covariance.

By using these lemmas we obtain the following proposition.

Proposition 4.3. Let $\mathbf{H}$ be a matrix in $\mathbb{F}_2^{r \times n}$ and let $\Gamma \in \mathbb{N}^{r \times r}$ be the intersection matrix of the rows of $\mathbf{H}$:

$$\Gamma_{ij} \stackrel{\text{def}}{=} |\text{supp}(\mathbf{h}_i) \cap \text{supp}(\mathbf{h}_j)|$$

where $i, j \in \llbracket 1, r \rrbracket$, $\mathbf{h}_i$ is the i -th row of $\mathbf{H}$ for $i \in \llbracket 1, r \rrbracket$. Let $\mathbf{e}$ be an element of $\mathbb{F}_2^n$ of weight t drawn uniformly at random. We have

$$\mathbf{Var}[S_{\text{BIKE}}] = r p_n(t, w)(1 - p_n(t, w)) + \sum_{i \neq j} \Sigma_n(\Gamma_{ij}, t, w).$$

We have performed an experimental study consisting in choosing random regular matrices $r \times 2r$ type ($w = 2d, d$) and random quasi-cyclic matrices of index 2 with circulant blocks of size r and weight d and have compared the distribution of $\mathbf{Var}[S_{\text{BIKE}}]$ in both cases for BIKE parameters. The difference between the quasi-cyclic case and the plain regular model is striking as shown in Figure 4.2. The mean of the variances are about the same but there is a huge difference in the variance of both distributions. In the plain regular case, the distribution of $\mathbf{Var}[S_{\text{BIKE}}]$ is much more concentrated around the mean than in the quasi-cyclic case.

The Case of HQC. The noisy syndrome weight S_{HQC} arising in HQC is of the form

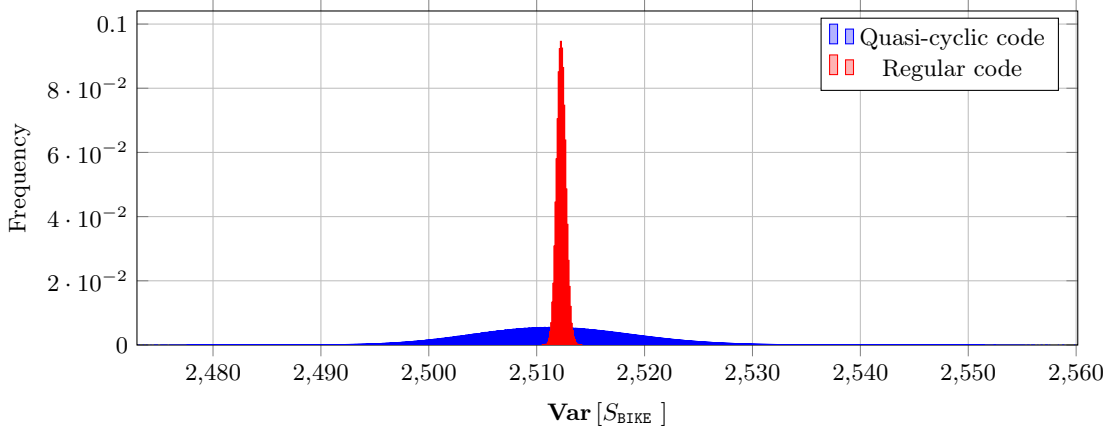
$$S_{\text{HQC}} = |\mathbf{r}_1 \mathbf{H}_1^\top + \mathbf{r}_2 \mathbf{H}_2^\top + \mathbf{e}| \quad (4.2)$$

Here $\mathbf{H}_1$ and $\mathbf{H}_2$ are circulant blocks of size r and constant column weight d , $\mathbf{r}_1$ and $\mathbf{r}_2$ are random errors of weight t_r and $\mathbf{e}$ is a random error of weight t_e that are all chosen uniformly at random.

In order to compute, for a given pair $(\mathbf{H}_1, \mathbf{H}_2)$, the variance $\mathbf{Var}[S_{\text{HQC}}]$ over $\mathbf{r}_1, \mathbf{r}_2$ and $\mathbf{e}$, we will need a few lemmas. It will be helpful, instead of working with bits b , to work with numbers $(-1)^b$ in $\{-1, 1\}$. This transforms the addition modulo 2 into a product. We relate the relevant quantities through the following observations

Lemma 4.4. Let X be two Bernoulli variables of parameter p , i.e. $\mathbf{P}(X = 1) = \mathbf{P}(Y = 1) = p$. Let $\tilde{X} \stackrel{\text{def}}{=} (-1)^X$ and $\tilde{Y} = (-1)^Y$. We have

$$\begin{aligned} \mathbb{E}(\tilde{X}) &= 1 - 2\mathbb{E}(X) = 1 - 2p \\ \mathbf{Var}[\tilde{X}] &= 4\mathbf{Var}[X] \\ \mathbf{Cov}(\tilde{X}, \tilde{Y}) &= 4\mathbf{Cov}(X, Y). \end{aligned}$$



Structure	N	Mean	Stddev
QC	631,000,000	2511.970242	7.366168
Reg	58,300	2511.984226	0.41935

Fig. 4.2: Distribution of the variance obtained using Proposition 4.3 for codes of parameters $(\text{id}\mathbf{x}, r, d, t) = (2, 12323, 71, 134)$ drawn uniformly at random. N denotes the number of samples we have drawn. QC refers to the quasi-cyclic case and Reg to the plain regular case. We have given the mean and standard deviations of $\text{Var}[S_{\text{BIKE}}]$ in both cases.

Lemma 4.5. *Let X_1, X_2, X_3, Y_1, Y_2 and Y_3 be six random variables taking their value in $\{-1, 1\}$ where all the pairs (X_i, Y_i) are independent random variables. We have*

$$\begin{aligned}
\text{Var}(X_1 X_2 X_3) &= \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) \\
&\quad - \text{Var}(X_1) \text{Var}(X_2) - \text{Var}(X_1) \text{Var}(X_3) - \text{Var}(X_2) \text{Var}(X_3) \\
&\quad + \text{Var}(X_1) \text{Var}(X_2) \text{Var}(X_3).
\end{aligned} \tag{4.3}$$

$$\begin{aligned}
\text{Cov}(X_1 X_2 X_3, Y_1 Y_2 Y_3) &= \overline{X_2} \cdot \overline{Y_2} \cdot \overline{X_3} \cdot \overline{Y_3} \text{Cov}(X_1, Y_1) \\
&\quad + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_3} \cdot \overline{Y_3} \text{Cov}(X_2, Y_2) \\
&\quad + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_2} \cdot \overline{Y_2} \text{Cov}(X_3, Y_3) \\
&\quad + \overline{X_3} \cdot \overline{Y_3} \text{Cov}(X_1, Y_1) \text{Cov}(X_2, Y_2) \\
&\quad + \overline{X_1} \cdot \overline{Y_1} \text{Cov}(X_2, Y_2) \text{Cov}(X_3, Y_3) \\
&\quad + \overline{X_2} \cdot \overline{Y_2} \text{Cov}(X_1, Y_1) \text{Cov}(X_3, Y_3) \\
&\quad + \text{Cov}(X_1, Y_1) \text{Cov}(X_2, Y_2) \text{Cov}(X_3, Y_3).
\end{aligned} \tag{4.4}$$

By putting all these results together we obtain the following proposition.

Proposition 4.6. *Let $\Gamma^{(1)}$ be the intersection matrix of $\mathbf{H}_1$ and $\Gamma^{(2)}$ be the intersection matrix of $\mathbf{H}_2$. Let t_r be the weight of $\mathbf{r}_1$ and $\mathbf{r}_2$ and t_e be the weight of $\mathbf{e}$. The row and column weights of $\mathbf{H}_1$ and $\mathbf{H}_2$ are all supposed to be of weight d . The length of $\mathbf{e}$, and the size of the square circulant matrices $\mathbf{H}_1$ and $\mathbf{H}_2$ are assumed to be equal to r . We also let*

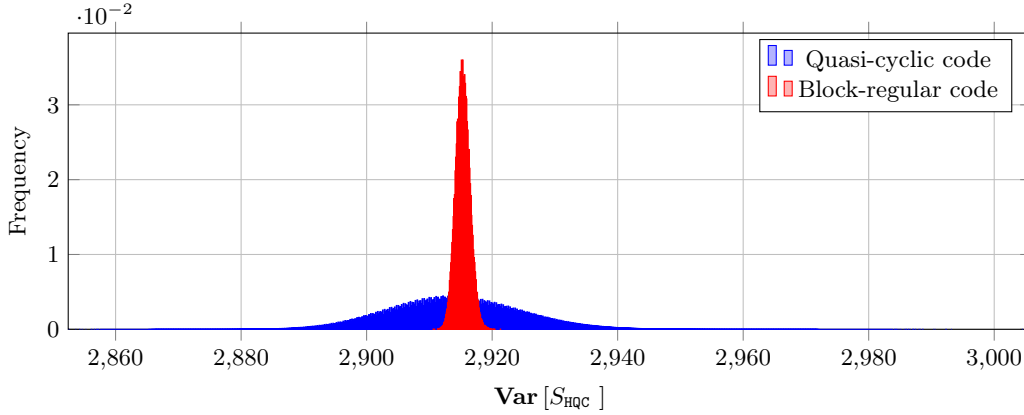
$$p \stackrel{\text{def}}{=} \mathbf{P}(\mathbf{r}_1 \mathbf{H}_1^\top(i) = 1), \quad q \stackrel{\text{def}}{=} \frac{t_e}{r}, \quad \alpha \stackrel{\text{def}}{=} \text{Cov}(e_i, e_j) \text{ for } i \neq j.$$

We have

$$\text{Var}[S_{\text{HQC}}] = rv + \sum_{i,j,i \neq j} c_{ij} \text{ where} \tag{4.5}$$

$$\begin{aligned}
v &\stackrel{\text{def}}{=} 2p(1-p) + q(1-q) - 4p^2(1-p)^2 - 8pq(1-p)(1-q) \\
&\quad + 16p^2q(1-p)^2(1-q) \\
c_{ij} &\stackrel{\text{def}}{=} \left\{ (1-2p)^2(1-2q)^2 + 4(1-2p)^2\alpha \right\} \left\{ \Sigma_r(\Gamma_{ij}^{(1)}, t_r, d) + \Sigma_r(\Gamma_{ij}^{(2)}, t_r, d) \right\} \\
&\quad + (1-2p)^4\alpha + (16\alpha + 4(1-2q)^2) \Sigma_r(\Gamma_{ij}^{(1)}, t_r, d) \cdot \Sigma_r(\Gamma_{ij}^{(2)}, t_r, d)
\end{aligned}$$

We have performed extensive experiments by computing $\text{Var}[S_{\text{HQC}}]$ for more than $2 \cdot 10^{10}$ different HQC keys drawn uniformly, and have compared the distribution we have obtained to the distribution of $\text{Var}[S_{\text{HQC}}]$ where $\mathbf{H}_1$ and $\mathbf{H}_2$ are just chosen as regular (d, d) square matrices of size r . We give the results in Figure 4.3. The first case is referred to as quasi-cyclic (QC) in the figure, and the second case as “block-regular”. Again the difference between both cases is striking. The variance of the quasi-cyclic case is tremendously larger than in the block-regular case. This raises the issue whether the fact that the variances $\text{Var}[S_{\text{HQC}}]$ of the quasi-cyclic HQC keys are largely spread does not lead to an excessively high DFR for weak HQC keys. We address this issue in the next subsection.



Structure	N	Mean	Var	Stddev	Min	Max
QC	22,975,000,000	2914.818874	112.068224	10.586228	2852.477113	3005.938514
Block-reg	59,800	2914.852920	1.298609	1.139565	2910.132253	2920.843229

Fig. 4.3: Distribution of the variance obtained using Proposition 4.6 for codes of parameters $(\text{idx}, r, d, s, t) = (2, 17669, 66, 75, 75)$ drawn uniformly at random. N denotes the number of samples we have drawn. QC refers to the quasi-cyclic case and Block-reg to the block regular case.

4.2 Analysis of the DFR of HQC.

We use these results to give an analysis of the DFR of HQC. In HQC, the decapsulation algorithm needs to successfully decode a noisy codeword in a code $\mathcal{C}$ which is a concatenated code using a shortened Reed-Solomon code as external code and a multiplicated Reed-Muller code as internal code. The error that has to be decoded is precisely the vector $\mathbf{e}_{\text{HQC}} \stackrel{\text{def}}{=} \mathbf{r}_1 \mathbf{H}_1^\top + \mathbf{r}_2 \mathbf{H}_2^\top + \mathbf{e}$ which appears in (4.2). In [AAB⁺22b], the failure probability of this decoding is computed by assuming that the error bits of $\mathbf{e}_{\text{HQC}}$ behave as independent Bernoulli random variables. In order to estimate the DFR more precisely from a refined model of the syndrome weight, we need to compute the failure probability for an error of a given weight. More precisely we will proceed as follows.

1. Use a normal approximation of the distribution of $S_{\text{HQC}} = |\mathbf{e}_{\text{HQC}}|$ based on its mean and variance. Let $p(t)$ be the probability that $S_{\text{HQC}} = t$.

2. Give a formula of the probability $\text{DFR}(t)$ that the concatenated code fails to decode $\mathbf{e}_{\text{HQC}}$ given that the weight S_{HQC} of $\mathbf{e}_{\text{HQC}}$ is equal to t .
3. Compute the DFR using the formula :

$$\text{DFR} = \sum_t p(t) \text{DFR}(t).$$

$\text{DFR}(t)$ is estimated through the following result.

Theorem 4.7 (DFR of the concatenated code). *The DFR of the concatenated code, using a $[n_e, k_e, d_e]_{\mathbb{F}_{256}}$ Reed-Solomon code as external code and an internal code of length n_i , for an error of weight t is :*

$$\text{DFR}(t) = \frac{1}{\binom{n}{t}} \sum_{\ell=\delta_e+1}^{n_e} a_{t,\ell}$$

with

$$\begin{aligned} a_{t,\ell} &= [X^t Y^\ell] \left(\left[\sum_{j=0}^{n_i} \binom{n_i}{j} (1 - \text{DFR}_i(j)) X^j + Y \cdot \sum_{j=0}^{n_i} \binom{n_i}{j} \text{DFR}_i(j) X^j \right]^{n_e} \right) \\ &= [X^t] \left(\binom{n_e}{\ell} \left[\sum_{j=0}^{n_i} \binom{n_i}{j} \text{DFR}_i(j) X^j \right]^\ell \left[\sum_{j=0}^{n_i} \binom{n_i}{j} (1 - \text{DFR}_i(j)) X^j \right]^{n_e - \ell} \right) \end{aligned}$$

where $n = n_e n_i$, $\delta_e = \lfloor \frac{d_e - 1}{2} \rfloor = \lfloor \frac{n_e - k_e}{2} \rfloor$ and, for $0 \leq j \leq n_i$, $\text{DFR}_i(j)$ is the DFR of the internal code for an error of weight j .

For the DFR of the internal code, we simulated 10^8 decodings for each error weight and we use the obtained experimental value when we observed sufficiently many failures, otherwise we use the following upper bound, adapted from [AAB⁺22b, Prop. 6.1.4] :

Proposition 4.8 (Upper bound on the DFR of the internal code). *The DFR of the $[n_i, k_i, d_i]_{\mathbb{F}_2}$ internal code for an error of weight j satisfy :*

- if $j < d_i/2$, $\text{DFR}_i(j) = 0$
- if $j \geq d_i/2$, we have the following upper bound :

$$\begin{aligned} \text{DFR}_i(j) &\leq \min \left(1, \frac{1}{\binom{n_i}{j}} \left[\frac{1}{2} 255 \binom{d_i}{d_i/2} \binom{d_i}{j - d_i/2} \right. \right. \\ &\quad \left. \left. + 255 \sum_{k=d_i/2+1}^{d_i} \binom{d_i}{k} \binom{d_i}{j - k} + \frac{1}{2} \binom{255}{2} \sum_{k=0}^{d_i/2} \binom{d_i/2}{k} \binom{d_i/2}{j - d_i + k} \right] \right) \end{aligned}$$

We compute $a_{t,\ell}$ using a saddlepoint approximation as described in §3.4. We have drawn more than $2 \cdot 10^{10}$ HQC keys, computed the relevant variances $\mathbf{Var}[S_{\text{HQC}}]$. Fig. 4.4 shows the smallest $\mathbf{Var}[S_{\text{HQC}}]$ (quasi-cyclic code 1) that was found together with the largest (quasi-cyclic code 6). We have computed the DFR for these two codes together with 4 other examples with variances ranging between these two extremes. The DFR ranges between 2^{-146} and 2^{-148} . This gives strong evidence that the DFR for HQC was estimated very conservatively. Now that we have a good understanding of the DFR of HQC we observe that there is some room for improving the parameters by still keeping a DFR of order $2^{-\lambda}$ where λ is the security parameter.

Acknowledgements

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Code	Theoretical variance	Experimental variance	$\log_2(\text{DFR})$
Best code	2852.477113	2852.11091474	-148.10969898
2nd best code	2853.032066	2852.82896851	-148.10329691
3rd best code	2854.164148	2854.51380444	-148.09023841
3rd worst code	2993.699413	2993.78639583	-146.49556243
2nd worst code	3004.997656	3004.84143353	-146.36771816
Worst code	3005.938514	3005.59234228	-146.35708052

Fig. 4.4: Lowest and highest theoretical variances of the syndrome weight among 2.10^{10} quasi-cyclic codes with HQC-1 parameters $(\text{idx}, r, d, t_e, t_r) = (2, 17669, 66, 75, 75)$, corresponding experimental variance obtained with 2.10^8 syndromes, and resulting DFR within the normal approximation.

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A Proofs for §3

A.1 Proofs for §3.1

We recall the corollaries that we are going to prove

Corollary 3.3. *Let w, d be fixed integers. For $r, n, t \in \mathbb{N}$ such that $rw = nd$, and $\mathbf{j} \in \llbracket 0, w \rrbracket^r$ such that $\sum_{i=1}^r j_i = dt$,*

$$\frac{|A_{r,t}^{\mathbf{j},d}| \cdot |A_{r,n-t}^{w-\mathbf{j},d}|}{|A_{r,n}^{w,d}|} \underset{n,t \rightarrow +\infty}{\sim} \frac{\prod_{i=1}^r \binom{w}{j_i} \exp\left(-\left[\frac{j_i(j_i-1)}{dt} + \frac{(w-j_i)(w-j_i-1)}{d(n-t)}\right] \frac{d-1}{2}\right)}{\binom{dn}{dt} \exp\left(-\frac{(w-1)(d-1)}{2}\right)}$$

Proof.

$$\begin{aligned} \frac{|A_{r,t}^{\mathbf{j},d}| \cdot |A_{r,n-t}^{w-\mathbf{j},d}|}{|A_{r,n}^{w,d}|} &\underset{n,t \rightarrow +\infty}{\sim} \frac{\frac{(dt)!}{\prod_{i=1}^r j_i! \cdot d!^t} \exp\left(-\frac{\sum_{i=1}^r j_i(j_i-1)}{dt} \frac{d-1}{2}\right) \cdot \frac{(d(n-t))!}{\prod_{i=1}^r (w-j_i)! \cdot d!^{n-t}} \exp\left(-\frac{\sum_{i=1}^r (w-j_i)(w-j_i-1)}{d(n-t)} \frac{d-1}{2}\right)}{\frac{(dn)!}{w!^r d!^n} \exp\left(-\frac{(w-1)(d-1)}{2}\right)} \\ &= \frac{\prod_{i=1}^r \frac{w!}{j_i!(w-j_i)!} \exp\left(-\frac{j_i(j_i-1)}{dt} \frac{d-1}{2}\right) \exp\left(-\frac{(w-j_i)(w-j_i-1)}{d(n-t)} \frac{d-1}{2}\right)}{\frac{(dn)!}{(dt)!(d(n-t))!} \exp\left(-\frac{(w-1)(d-1)}{2}\right)} \\ &= \frac{\prod_{i=1}^r \binom{w}{j_i} \exp\left(-\left[\frac{j_i(j_i-1)}{dt} + \frac{(w-j_i)(w-j_i-1)}{d(n-t)}\right] \frac{d-1}{2}\right)}{\binom{dn}{dt} \exp\left(-\frac{(w-1)(d-1)}{2}\right)} \end{aligned}$$

□

Corollary 3.4. *Let w, d be fixed integers. For $r, n, t \in \mathbb{N}$, and $\mathbf{w}, \mathbf{j} \in \llbracket 0, w \rrbracket^r$ such that $0 \leq j_i \leq w_i$ for $1 \leq i \leq r$, $\sum_{i=1}^r w_i = dn$ and $\sum_{i=1}^r j_i = dt$,*

$$\frac{|A_{r,t}^{\mathbf{j},d}| \cdot |A_{r,n-t}^{\mathbf{w}-\mathbf{j},d}|}{|A_{r,n}^{w,d}|} \underset{n,t \rightarrow +\infty}{\sim} \frac{\prod_{i=1}^r \binom{w_i}{j_i} \exp\left(-\left[\frac{j_i(j_i-1)}{dt} + \frac{(w_i-j_i)(w_i-j_i-1)}{d(n-t)}\right] \frac{d-1}{2}\right)}{\binom{dn}{dt} \exp\left(-\frac{\sum_{i=1}^r w_i(w_i-1)}{dn} \frac{d-1}{2}\right)}$$

Proof.

$$\begin{aligned} \frac{|A_{r,t}^{\mathbf{j},d}| \cdot |A_{r,n-t}^{w-\mathbf{j},d}|}{|A_{r,n}^{w,d}|} &\underset{n,t \rightarrow +\infty}{\sim} \frac{\frac{(dt)!}{\prod_{i=1}^r j_i! \cdot d!^t} \exp\left(-\frac{\sum_{i=1}^r j_i(j_i-1)}{dt} \frac{d-1}{2}\right) \cdot \frac{(d(n-t))!}{\prod_{i=1}^r (w_i-j_i)! \cdot d!^{n-t}} \exp\left(-\frac{\sum_{i=1}^r (w_i-j_i)(w_i-j_i-1)}{d(n-t)} \frac{d-1}{2}\right)}{\frac{(dn)!}{w!^r d!^n} \exp\left(-\frac{\sum_{i=1}^r w_i(w_i-1)}{dn} \frac{d-1}{2}\right)} \\ &= \frac{\prod_{i=1}^r \frac{w_i!}{j_i!(w_i-j_i)!} \exp\left(-\frac{j_i(j_i-1)}{dt} \frac{d-1}{2}\right) \exp\left(-\frac{(w_i-j_i)(w_i-j_i-1)}{d(n-t)} \frac{d-1}{2}\right)}{\frac{(dn)!}{(dt)!(d(n-t))!} \exp\left(-\frac{\sum_{i=1}^r w_i(w_i-1)}{dn} \frac{d-1}{2}\right)} \\ &= \frac{\prod_{i=1}^r \binom{w_i}{j_i} \exp\left(-\left[\frac{j_i(j_i-1)}{dt} + \frac{(w_i-j_i)(w_i-j_i-1)}{d(n-t)}\right] \frac{d-1}{2}\right)}{\binom{dn}{dt} \exp\left(-\frac{\sum_{i=1}^r w_i(w_i-1)}{dn} \frac{d-1}{2}\right)} \end{aligned}$$

A.2 Proofs for §3.2

We prove here the following theorem

Theorem 3.5. *Let w, d be fixed integers. Let $r, n, t \in \mathbb{N}$ be such that $rw = nd$. Let $\mathbf{H}$ and $\mathbf{e}$ be random variables uniformly distributed over $\Lambda_{r,n}^{w,d}$ and the set of binary vectors of length n and Hamming weight t respectively. Let $S = |\mathbf{eH}^\top|$. If $s - dt \equiv 1 \pmod{2}$, then $\mathbf{P}(S = s) = 0$ and otherwise we have*

$$\mathbf{P}(S = s) \underset{n,t \rightarrow +\infty}{\sim} \frac{b_{dt,s}^{(n,w,d,t)}}{\binom{dn}{dt} \exp\left(-\frac{(w-1)(d-1)}{2}\right)}$$

where

$$\begin{aligned} b_{j,s}^{(n,w,d,t)} &\stackrel{\text{def}}{=} [X^j Y^s] \left(\left[f_{\text{even}}^{(n,w,d,t)}(X) + Y \cdot f_{\text{odd}}^{(n,w,d,t)}(X) \right]^r \right) \\ &= [X^j] \left(\binom{r}{s} \left[f_{\text{odd}}^{(n,w,d,t)}(X) \right]^s \left[f_{\text{even}}^{(n,w,d,t)}(X) \right]^{r-s} \right). \end{aligned}$$

Proof. Let $\mathbf{e}_t$ be the vector with 1's on the first t positions, and 0's on the remaining $n-t$ positions.

For any $\mathbf{e} \in \mathbb{F}_2^n$ such that $|\mathbf{e}| = t$, there exists a permutation matrix $\mathbf{P}$ such that $\mathbf{eP} = \mathbf{e}_t$, and we have $\mathbf{eH}^\top = \mathbf{ePP}^\top \mathbf{H}^\top = \mathbf{e}_t(\mathbf{HP})^\top$. Since for any invertible matrix $\mathbf{P}$, $\mathbf{H} \mapsto \mathbf{HP}$ is a permutation of $\Lambda_{r,n}^{w,d}$, $\mathbf{HP}$ is uniformly distributed over $\Lambda_{r,n}^{w,d}$. Thus, $\mathbf{P}(|\mathbf{eH}^\top| = s) = \mathbf{P}(|\mathbf{e}_t \mathbf{H}^\top| = s)$.

Therefore,

$$\mathbf{P}(S = s) = \frac{|\Lambda_{r,n}^{w,d}(t, s)|}{|\Lambda_{r,n}^{w,d}|}$$

where $\Lambda_{r,n}^{w,d}(t, s) \stackrel{\text{def}}{=} \{\mathbf{H} \in \Lambda_{r,n}^{w,d} \mid |\mathbf{e}_t \mathbf{H}^\top| = s\}$.

Let $\mathbf{H} \in \Lambda_{r,n}^{w,d}$. For $1 \leq i \leq r$, let j_i be the weight of the i^{th} row restricted to the t positions of the error. Note that $0 \leq j_i \leq w$ for all $i \in [1, r]$ and $\sum_{0 \leq i \leq r} j_i = dt$. We have that $\mathbf{H} \in \Lambda_{r,n}^{w,d}(t, s)$ if and only if there are exactly s rows such that j_i is odd and $r-s$ rows such that j_i is even. Thus,

$$|\Lambda_{r,n}^{w,d}(t, s)| = \sum_{\substack{0 \leq \mathbf{j} \leq w \\ j_1 + \dots + j_r = dt \\ |\mathbf{j} \cap (2\mathbb{N}+1)| = s}} |\Lambda_{r,t}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-t}^{w-\mathbf{j},d}|$$

and therefore,

$$\begin{aligned} \mathbf{P}(S = s) &= \sum_{\substack{0 \leq \mathbf{j} \leq w \\ j_1 + \dots + j_r = dt \\ |\mathbf{j} \cap (2\mathbb{N}+1)| = s}} \frac{|\Lambda_{r,t}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-t}^{w-\mathbf{j},d}|}{|\Lambda_{r,n}^{w,d}|} \\ &\underset{n,t \rightarrow +\infty}{\sim} \sum_{\substack{0 \leq \mathbf{j} \leq w \\ j_1 + \dots + j_r = dt \\ |\mathbf{j} \cap (2\mathbb{N}+1)| = s}} \frac{\prod_{i=1}^r a_{j_i}^{(n,w,d,t)}}{\binom{dn}{dt} \exp\left(-\frac{(w-1)(d-1)}{2}\right)} \text{ by Corollary 3.3} \\ &= \frac{b_{dt,s}^{(n,w,d,t)}}{\binom{dn}{dt} \exp\left(-\frac{(w-1)(d-1)}{2}\right)} \end{aligned}$$

□

A.3 Proof of Theorem 3.8

Recall this theorem

Theorem 3.8. Let $\text{idx}, r, d, t, u, s$ be integers such that $0 \leq u \leq \min(d, t)$ and $0 \leq s \leq dt - u(u-1)$, $w = \text{idx} \cdot d$. Let $\Delta \in \llbracket 0, d \rrbracket^r$. Let $\mathbf{H}$ be a random variable uniformly distributed over the set of matrices formed by concatenating side by side idx circulant matrices of size r and weight d that correspond to quasi-cyclic code that have a near codeword ν whose degree sequence $\Delta(\nu)$ satisfies $\Delta(\nu) = \Delta$. Let $\mathbf{e}$ be a random variable uniformly distributed the set of binary vectors of length n and weight t . Let $U = |\nu \cap \mathbf{e}|$ and $S = |\mathbf{e}\mathbf{H}^\top|$. If $s - dt \equiv 1 \pmod{2}$, then $\mathbf{P}(S = s \mid U = u) = 0$ and otherwise under Assumption 2 we have

$$\mathbf{P}(S = s \mid U = u) = \sum_{\substack{0 \leq \mathbf{k} \leq \Delta \\ 0 \leq \mathbf{j} \leq w - \Delta \\ k_1 + \dots + k_r = du \\ j_1 + \dots + j_r = d(t-u) \\ |(\mathbf{k} + \mathbf{j}) \cap (2\mathbb{N} + 1)| = s}} \frac{|\Lambda_{r,u}^{\mathbf{k},d}| \cdot |\Lambda_{r,d-u}^{\Delta - \mathbf{k},d}|}{|\Lambda_{r,d}^{\Delta,d}|} \cdot \frac{|\Lambda_{r,t-u}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-d-t+u}^{w-\Delta-\mathbf{j},d}|}{|\Lambda_{r,n-d}^{w-\Delta,d}|}$$

Proof. Let $\mathbf{e}_u \in \mathbb{F}_2^d$ be the vector with 1's on the first u positions, and 0's on the remaining $d - u$ positions. Let $\mathbf{e}_{t-u} \in \mathbb{F}_2^{n-d}$ be the vector with 1's on the first $t - u$ positions, and 0's on the remaining $n - d - t + u$ positions. Let $\mathbf{e}_{t,u} = (\mathbf{e}_u \mathbf{e}_{t-u}) \in \mathbb{F}_2^n$. We also let $\nu' \stackrel{\text{def}}{=} \mathbf{e}_{t,u}$.

For any $(\mathbf{e}_0, \mathbf{e}_1) \in \mathbb{F}_2^d \times \mathbb{F}_2^{n-d}$ such that $|\mathbf{e}_0| = u$ and $|\mathbf{e}_1| = t - u$, there exists two permutation matrices $\mathbf{P}_0$ and $\mathbf{P}_1$ such that $\mathbf{e}_0 \mathbf{P}_0 = \mathbf{e}_u$ and $\mathbf{e}_1 \mathbf{P}_1 = \mathbf{e}_{t-u}$. For $e = (\mathbf{e}_0 \mathbf{e}_1)$ and $\mathbf{P} = \begin{pmatrix} \mathbf{P}_0 & \mathbf{0} \\ \mathbf{0} & \mathbf{P}_1 \end{pmatrix}$, we have that $\mathbf{e}\mathbf{H}^\top = \mathbf{e}\mathbf{P}\mathbf{P}^\top\mathbf{H}^\top = \mathbf{e}_{t,u}(\mathbf{H}'\mathbf{P})^\top$. Since $\mathbf{H}'_0$ is uniformly distributed over $\Lambda_{r,d}^{\Delta,d}$ and, for any invertible matrix $\mathbf{P}$, $\mathbf{H}' \mapsto \mathbf{H}'\mathbf{P}$ is a permutation of $\Lambda_{r,d}^{\Delta,d}$, $\mathbf{H}'_0 \mathbf{P}_0$ is uniformly distributed over $\Lambda_{r,d}^{\Delta,d}$. Similarly, $\mathbf{H}'_1 \mathbf{P}_1$ is uniformly distributed over $\Lambda_{r,n-d}^{w-\Delta,d}$.

Therefore,

$$\mathbf{P}(S' = s \mid U' = u) = \frac{|\Gamma_{r,n}^{w,d,\Delta}(t, u, s)|}{|\Gamma_{r,n}^{w,d,\Delta}|}$$

where $\Gamma_{r,n}^{w,d,\Delta} = \Lambda_{r,d}^{\Delta,d} \times \Lambda_{r,n-d}^{w-\Delta,d}$ and $\Gamma_{r,n}^{w,d,\Delta}(t, u, s) = \{\mathbf{H}' \in \Gamma_{r,n}^{w,d,\Delta} \mid |\mathbf{e}_{t,u}\mathbf{H}'^\top| = s\}$

Let $\mathbf{H}' \in \Gamma_{r,n}^{w,d,\Delta}$. For $1 \leq i \leq r$, let k_i (resp. j_i) be the weight of the i^{th} row of $\mathbf{H}'$ restricted to the u (resp. $t - u$) positions in the support of $\mathbf{e}_u$ (resp. $\mathbf{e}_{t-u}$). We have that :

- $\forall 1 \leq i \leq r, 0 \leq k_i \leq \Delta_i$
- $\forall 1 \leq i \leq r, 0 \leq j_i \leq w - \Delta_i$
- $\sum_{1 \leq i \leq r} k_i = du$
- $\sum_{1 \leq i \leq r} j_i = d(t - u)$
- $\mathbf{H}' \in \Gamma_{r,n}^{w,d,\Delta}(t, u, s)$ if and only if there are s indexes i such that $k_i + j_i$ is odd and $r - s$ such that $k_i + j_i$ is even

Thus,

$$|\Gamma_{r,n}^{w,d,\Delta}(t, u, s)| = \sum_{\substack{0 \leq \mathbf{k} \leq \Delta \\ 0 \leq \mathbf{j} \leq w - \Delta \\ k_1 + \dots + k_r = du \\ j_1 + \dots + j_r = d(t-u) \\ |(\mathbf{k} + \mathbf{j}) \cap (2\mathbb{N} + 1)| = s}} |\Lambda_{r,u}^{\mathbf{k},d}| \cdot |\Lambda_{r,d-u}^{\Delta - \mathbf{k},d}| \cdot |\Lambda_{r,t-u}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-d-t+u}^{w-\Delta-\mathbf{j},d}|$$

and therefore

$$\mathbf{P}(S' = s \mid U' = u) = \sum_{\substack{0 \leq \mathbf{k} \leq \Delta \\ 0 \leq \mathbf{j} \leq w - \Delta \\ k_1 + \dots + k_r = du \\ j_1 + \dots + j_r = d(t-u) \\ |(\mathbf{k} + \mathbf{j}) \cap (2\mathbb{N} + 1)| = s}} \frac{|\Lambda_{r,u}^{\mathbf{k},d}| \cdot |\Lambda_{r,d-u}^{\Delta - \mathbf{k},d}|}{|\Lambda_{r,d}^{\Delta,d}|} \cdot \frac{|\Lambda_{r,t-u}^{\mathbf{j},d}| \cdot |\Lambda_{r,n-d-t+u}^{w-\Delta-\mathbf{j},d}|}{|\Lambda_{r,n-d}^{w-\Delta,d}|}$$

B Proofs of §4

B.1 Proofs for §4.1

Let us recall Lemma 4.1

Lemma 4.1. *Let $\mathbf{h} \in \mathbb{F}_2^n$ of Hamming weight w and assume that $\mathbf{e}$ is an error of Hamming weight t in $\mathbb{F}_2^n$ chosen uniformly at random. Let $p_n(t, w) \stackrel{\text{def}}{=} \mathbf{P}(\langle \mathbf{h}, \mathbf{e} \rangle = 1)$. We have*

$$p_n(t, w) = \sum_{\substack{i=1 \\ i \text{ odd}}}^n \frac{\binom{w}{i} \binom{n-w}{t-i}}{\binom{n}{t}}.$$

Proof. We clearly have

$$\begin{aligned} \mathbf{P}(\langle \mathbf{h}, \mathbf{e} \rangle = 1) &= \sum_{\substack{i=1 \\ i \text{ odd}}}^n \mathbf{P}(|\mathbf{h} \cap \mathbf{e}| = i) \\ &= \frac{\binom{w}{i} \binom{n-w}{t-i}}{\binom{n}{t}}. \end{aligned}$$

□

We also prove here

Lemma 4.2. *Let $\mathbf{h}$ and $\mathbf{h}'$ be two vectors of $\mathbb{F}_2^n$ of weight w . Let $a \stackrel{\text{def}}{=} |\mathbf{h} \cap \mathbf{h}'|$. Let $\mathbf{e}$ be an element of $\mathbb{F}_2^n$ of weight t drawn uniformly at random. We have*

$$\text{Cov}(\langle \mathbf{h}, \mathbf{e} \rangle, \langle \mathbf{h}', \mathbf{e} \rangle) = \sum_{\substack{p, q, r, s \in \mathbb{N} \\ p+q \text{ odd} \\ p+r \text{ odd} \\ p+q+r+s=t}} \frac{\binom{a}{p} \binom{w-a}{q} \binom{w-a}{r} \binom{n-2w+a}{s}}{\binom{n}{t}} - p_n(t, w)^2.$$

We denote by $\Sigma_n(a, t, w)$ this expression for the covariance.

Proof. We have

$$\text{Cov}(\langle \mathbf{h}, \mathbf{e} \rangle, \langle \mathbf{h}', \mathbf{e} \rangle) = \mathbf{P}(\langle \mathbf{h}, \mathbf{e} \rangle = 1, \langle \mathbf{h}', \mathbf{e} \rangle = 1) - \mathbf{P}(\langle \mathbf{h}, \mathbf{e} \rangle = 1)^2. \quad (\text{B.1})$$

Notice that

$$\mathbf{P}(\langle \mathbf{h}, \mathbf{e} \rangle = 1, \langle \mathbf{h}', \mathbf{e} \rangle = 1) = \sum_{\substack{p, q, r, s \in \mathbb{N} \\ p+q \text{ odd} \\ p+r \text{ odd} \\ p+q+r+s=t}} P(p, q, r, s) \quad (\text{B.2})$$

where $P(p, q, r, s)$ is the probability that $\mathbf{e}$ intersects $\mathbf{h} \cap \mathbf{h}'$ in exactly p positions, $\mathbf{h} \setminus \mathbf{h}'$ in exactly q positions, $\mathbf{h}' \setminus \mathbf{h}$ in exactly r positions. We clearly have

$$P(p, q, r, s) = \frac{\binom{a}{p} \binom{w-a}{q} \binom{w-a}{r} \binom{n-2w+a}{s}}{\binom{n}{t}}.$$

Plugging this expression in (B.2), then into (B.1) and using Lemma 4.1 yields our result. □

These lemmas are used to prove

Proposition 4.3. Let $\mathbf{H}$ be a matrix in $\mathbb{F}_2^{r \times n}$ and let $\Gamma \in \mathbb{N}^{r \times r}$ be the intersection matrix of the rows of $\mathbf{H}$:

$$\Gamma_{ij} \stackrel{\text{def}}{=} |\text{supp}(\mathbf{h}_i) \cap \text{supp}(\mathbf{h}_j)|$$

where $i, j \in \llbracket 1, r \rrbracket$, $\mathbf{h}_i$ is the i -th row of $\mathbf{H}$ for $i \in \llbracket 1, r \rrbracket$. Let $\mathbf{e}$ be an element of $\mathbb{F}_2^n$ of weight t drawn uniformly at random. We have

$$\mathbf{Var}[S_{\text{BIKE}}] = rp_n(t, w)(1 - p_n(t, w)) + \sum_{i \neq j} \Sigma_n(\Gamma_{ij}, t, w).$$

Proof. We write S as

$$S = \sum_{i=1}^r X_i \quad \text{where} \\ X_i \stackrel{\text{def}}{=} \langle \mathbf{h}_i, \mathbf{e} \rangle.$$

By using this expression as a sum we have

$$\mathbf{Var}[S] = \sum_{i=1}^r \mathbf{Var}[X_i] + \sum_{i \neq j} \mathbf{Cov}(X_i, X_j). \quad (\text{B.3})$$

We clearly have $\mathbf{Var}[X_i] = p_n(t, w)(1 - p_n(t, w))$ by using Lemma 4.1 and $\mathbf{Cov}(X_i, X_j) = \Sigma_n(\Gamma_{ij}, t, w)$ by using Lemma 4.2. This finishes the proof by plugging these expressions into (B.3).

Let us first recall Lemma 4.5

Lemma 4.5. Let X_1, X_2, X_3, Y_1, Y_2 and Y_3 be six random variables taking their value in $\{-1, 1\}$ where all the pairs (X_i, Y_i) are independent random variables. We have

$$\begin{aligned} \mathbf{Var}(X_1 X_2 X_3) &= \mathbf{Var}(X_1) + \mathbf{Var}(X_2) + \mathbf{Var}(X_3) \\ &\quad - \mathbf{Var}(X_1) \mathbf{Var}(X_2) - \mathbf{Var}(X_1) \mathbf{Var}(X_3) - \mathbf{Var}(X_2) \mathbf{Var}(X_3) \\ &\quad + \mathbf{Var}(X_1) \mathbf{Var}(X_2) \mathbf{Var}(X_3). \end{aligned} \quad (4.3)$$

$$\begin{aligned} \mathbf{Cov}(X_1 X_2 X_3, Y_1 Y_2 Y_3) &= \overline{X_2} \cdot \overline{Y_2} \cdot \overline{X_3} \cdot \overline{Y_3} \mathbf{Cov}(X_1, Y_1) \\ &\quad + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_3} \cdot \overline{Y_3} \mathbf{Cov}(X_2, Y_2) \\ &\quad + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_3, Y_3) \\ &\quad + \overline{X_3} \cdot \overline{Y_3} \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_2, Y_2) \\ &\quad + \overline{X_1} \cdot \overline{Y_1} \mathbf{Cov}(X_2, Y_2) \mathbf{Cov}(X_3, Y_3) \\ &\quad + \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_3, Y_3) \\ &\quad + \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_2, Y_2) \mathbf{Cov}(X_3, Y_3). \end{aligned} \quad (4.4)$$

Proof. Let us first compute $\mathbf{Var}(X_1 X_2)$. We have

$$\begin{aligned} \mathbf{Var}(X_1 X_2) &= \overline{X_1^2 X_2^2} - \overline{X_1 X_2}^2 \\ &= \overline{X_1^2} \cdot \overline{X_2^2} - \overline{X_1}^2 \cdot \overline{X_2}^2 \\ &= \overline{X_1^2} \cdot \overline{X_2^2} - \overline{X_1^2} \cdot \overline{X_2}^2 + \overline{X_1^2} \cdot \overline{X_2}^2 - \overline{X_1}^2 \cdot \overline{X_2}^2 \\ &= \overline{X_1^2} \mathbf{Var}[X_2] + \overline{X_2}^2 \mathbf{Var}[X_1] \\ &= \mathbf{Var}[X_2] + (1 - \mathbf{Var}[X_2]) \mathbf{Var}[X_1] \\ &= \mathbf{Var}[X_1] + \mathbf{Var}[X_2] - \mathbf{Var}[X_1] \mathbf{Var}[X_2]. \end{aligned}$$

From this we deduce

$$\begin{aligned}
\mathbf{Var}(X_1 X_2 X_3) &= \mathbf{Var}[X_1 X_2] + \mathbf{Var}[X_3] - \mathbf{Var}[X_1 X_2] \mathbf{Var}[X_3] \\
&= \mathbf{Var}[X_1] + \mathbf{Var}[X_2] - \mathbf{Var}[X_1] \mathbf{Var}[X_2] + \mathbf{Var}[X_3] \\
&\quad - (\mathbf{Var}[X_1] + \mathbf{Var}[X_2] - \mathbf{Var}[X_1] \mathbf{Var}[X_2]) \mathbf{Var}[X_3] \\
&= \mathbf{Var}(X_1) + \mathbf{Var}(X_2) + \mathbf{Var}(X_3) - \mathbf{Var}(X_1) \mathbf{Var}(X_2) - \mathbf{Var}(X_1) \mathbf{Var}(X_3) \\
&\quad - \mathbf{Var}(X_2) \mathbf{Var}(X_3) + \mathbf{Var}(X_1) \mathbf{Var}(X_2) \mathbf{Var}(X_3).
\end{aligned}$$

For the covariance formula we first compute $\mathbf{Cov}(X_1, X_2)$

$$\begin{aligned}
\mathbf{Cov}(X_1 X_2, Y_1 Y_2) &= \overline{X_1 Y_1 X_2 Y_2} - \overline{X_1 X_2} \cdot \overline{Y_1 Y_2} \\
&= \overline{X_1 Y_1} \cdot \overline{X_2 Y_2} - \overline{X_1} \cdot \overline{X_2} \cdot \overline{Y_1} \cdot \overline{Y_2} \\
&= \overline{X_1 Y_1} \cdot \overline{X_2 Y_2} - \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_2 Y_2} + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_2 Y_2} - \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_2} \cdot \overline{Y_2} \\
&= \overline{X_2 Y_2} \mathbf{Cov}(X_1, Y_1) + \overline{X_1} \cdot \overline{Y_1} \mathbf{Cov}(X_2, Y_2) \\
&= \{\overline{X_2} \cdot \overline{Y_2} + \mathbf{Cov}(X_2, Y_2)\} \mathbf{Cov}(X_1, Y_1) + \overline{X_1} \cdot \overline{Y_1} \mathbf{Cov}(X_2, Y_2) \\
&= \overline{X_1} \cdot \overline{Y_1} \mathbf{Cov}(X_2, Y_2) + \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_1, Y_1) + \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_2, Y_2).
\end{aligned}$$

We use this in the computation of $C \stackrel{\text{def}}{=} \mathbf{Cov}(X_1 X_2 X_3, Y_1 Y_2 Y_3)$ as follows

$$\begin{aligned}
C &= \overline{X_3} \cdot \overline{Y_3} \mathbf{Cov}(X_1 X_2, Y_1 Y_2) + \overline{X_1 X_2} \cdot \overline{Y_1 Y_2} \mathbf{Cov}(X_3, Y_3) + \mathbf{Cov}(X_1 X_2, Y_1 Y_2) \mathbf{Cov}(X_3, Y_3) \\
&= \overline{X_3} \cdot \overline{Y_3} \{\overline{X_1} \cdot \overline{Y_1} \mathbf{Cov}(X_2, Y_2) + \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_1, Y_1) + \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_2, Y_2)\} \\
&\quad + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_3, Y_3) \\
&\quad + \{\overline{X_1} \cdot \overline{Y_1} \mathbf{Cov}(X_2, Y_2) + \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_1, Y_1) + \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_2, Y_2)\} \mathbf{Cov}(X_3, Y_3) \\
&= \overline{X_2} \cdot \overline{Y_2} \cdot \overline{X_3} \cdot \overline{Y_3} \mathbf{Cov}(X_1, Y_1) + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_3} \cdot \overline{Y_3} \mathbf{Cov}(X_2, Y_2) + \overline{X_1} \cdot \overline{Y_1} \cdot \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_3, Y_3) + \\
&\quad \overline{X_3} \cdot \overline{Y_3} \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_2, Y_2) + \overline{X_1} \cdot \overline{Y_1} \mathbf{Cov}(X_2, Y_2) \mathbf{Cov}(X_3, Y_3) + \overline{X_2} \cdot \overline{Y_2} \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_3, Y_3) \\
&\quad + \mathbf{Cov}(X_1, Y_1) \mathbf{Cov}(X_2, Y_2) \mathbf{Cov}(X_3, Y_3)
\end{aligned}$$

□

This Lemma is used to prove Proposition 4.6 which we now recall

Proposition 4.6. *Let $\Gamma^{(1)}$ be the intersection matrix of $\mathbf{H}_1$ and $\Gamma^{(2)}$ be the intersection matrix of $\mathbf{H}_2$. Let t_r be the weight of $\mathbf{r}_1$ and $\mathbf{r}_2$ and t_e be the weight of $\mathbf{e}$. The row and column weights of $\mathbf{H}_1$ and $\mathbf{H}_2$ are all supposed to be of weight d . The length of $\mathbf{e}$, and the size of the square circulant matrices $\mathbf{H}_1$ and $\mathbf{H}_2$ are assumed to be equal to r . We also let*

$$p \stackrel{\text{def}}{=} \mathbf{P}(\mathbf{r}_1 \mathbf{H}_1^\top(i) = 1), \quad q \stackrel{\text{def}}{=} \frac{t_e}{r}, \quad \alpha \stackrel{\text{def}}{=} \mathbf{Cov}(e_i, e_j) \text{ for } i \neq j.$$

We have

$$\mathbf{Var}[S_{\text{HQC}}] = rv + \sum_{i,j,i \neq j} c_{ij} \text{ where} \quad (4.5)$$

$$\begin{aligned}
v &\stackrel{\text{def}}{=} 2p(1-p) + q(1-q) - 4p^2(1-p)^2 - 8pq(1-p)(1-q) \\
&\quad + 16p^2q(1-p)^2(1-q) \\
c_{ij} &\stackrel{\text{def}}{=} \{(1-2p)^2(1-2q)^2 + 4(1-2p)^2\alpha\} \left\{ \Sigma_r(\Gamma_{ij}^{(1)}, t_r, d) + \Sigma_r(\Gamma_{ij}^{(2)}, t_r, d) \right\} \\
&\quad + (1-2p)^4\alpha + (16\alpha + 4(1-2q)^2) \Sigma_r(\Gamma_{ij}^{(1)}, t_r, d) \cdot \Sigma_r(\Gamma_{ij}^{(2)}, t_r, d)
\end{aligned}$$

Proof. We choose two arbitrary different positions i and j in $\llbracket 1, r \rrbracket$ and let

$$\begin{aligned} X_1 &\stackrel{\text{def}}{=} \mathbf{r}_1 \mathbf{H}_1^\top(i) \\ X_2 &\stackrel{\text{def}}{=} \mathbf{r}_2 \mathbf{H}_2^\top(i) \\ X_3 &\stackrel{\text{def}}{=} \mathbf{e}_i \\ Y_1 &\stackrel{\text{def}}{=} \mathbf{r}_1 \mathbf{H}_1^\top(j) \\ Y_2 &\stackrel{\text{def}}{=} \mathbf{r}_2 \mathbf{H}_2^\top(j) \\ Y_3 &\stackrel{\text{def}}{=} \mathbf{e}_j, \end{aligned}$$

and we also define the following associated random variables taking their values in $\{-1, 1\}$ as $\tilde{X}_k = (-1)^{X_k}$ and $\tilde{Y}_k = (-1)^{Y_k}$ for $k \in \llbracket 1, 3 \rrbracket$. Let

$$\begin{aligned} S &\stackrel{\text{def}}{=} X_1 \oplus X_2 \oplus X_3 \\ T &\stackrel{\text{def}}{=} Y_1 \oplus Y_2 \oplus Y_3 \end{aligned}$$

and $\tilde{S} = (-1)^S$ and $\tilde{T} = (-1)^T$. Observe that

$$\tilde{S} = \tilde{X}_1 \cdot \tilde{X}_2 \cdot \tilde{X}_3, \quad \tilde{T} = \tilde{Y}_1 \cdot \tilde{Y}_2 \cdot \tilde{Y}_3.$$

We also know by Lemma 4.4 that

$$\mathbf{Var}[S] = \frac{1}{4} \mathbf{Var}[\tilde{S}] \tag{B.4}$$

$$\mathbf{Cov}(S, T) = \frac{1}{4} \mathbf{Cov}(\tilde{S}, \tilde{T}). \tag{B.5}$$

By using this remark we obtain

$$\begin{aligned} \mathbf{Var}[S] &= \frac{1}{4} \mathbf{Var}[\tilde{S}] \\ &= \frac{1}{4} \mathbf{Var}[\tilde{X}_1 \tilde{X}_2 \tilde{X}_3] \\ &= \frac{1}{4} \left\{ \mathbf{Var}[\tilde{X}_1] + \mathbf{Var}[\tilde{X}_2] + \mathbf{Var}[\tilde{X}_3] - \mathbf{Var}[\tilde{X}_1] \mathbf{Var}[\tilde{X}_2] - \mathbf{Var}[\tilde{X}_1] \mathbf{Var}[\tilde{X}_3] \right. \\ &\quad \left. - \mathbf{Var}[\tilde{X}_2] \mathbf{Var}[\tilde{X}_3] + \mathbf{Var}[\tilde{X}_1] \mathbf{Var}[\tilde{X}_2] \mathbf{Var}[\tilde{X}_3] \right\}. \end{aligned}$$

We have

$$\begin{aligned} \mathbf{Var}[\tilde{X}_1] &= \mathbf{Var}[\tilde{X}_2] = 4p(1-p) \\ \mathbf{Var}[\tilde{X}_3] &= 4\mathbf{Var}[X_3] = 4q(1-q). \end{aligned}$$

From this we deduce that

$$\begin{aligned} \mathbf{Var}[S] &= \frac{1}{4} \{ 8p(1-p) + 4q(1-q) - 16p^2(1-p)^2 - 32pq(1-p)(1-q) + 64p^2q(1-p)^2(1-q) \} \\ &= 2p(1-p) + q(1-q) - 4p^2(1-p)^2 - 8pq(1-p)(1-q) + 16p^2q(1-p)^2(1-q). \end{aligned}$$

The covariance is treated similarly by using that

$$\begin{aligned} \mathbf{Cov}(S, T) &= \frac{1}{4} \mathbf{Cov}(\tilde{S}, \tilde{T}) \\ &= \frac{1}{4} \mathbf{Cov}(\tilde{X}_1 \tilde{X}_2 \tilde{X}_3, \tilde{Y}_1 \tilde{Y}_2 \tilde{Y}_3) \\ &= \frac{1}{4} \left\{ \overline{\tilde{X}_2 \cdot \tilde{Y}_2} \cdot \overline{\tilde{X}_3 \cdot \tilde{Y}_3} \mathbf{Cov}(\tilde{X}_1, \tilde{Y}_1) + \overline{\tilde{X}_1 \cdot \tilde{Y}_1} \cdot \overline{\tilde{X}_3 \cdot \tilde{Y}_3} \mathbf{Cov}(\tilde{X}_2, \tilde{Y}_2) + \overline{\tilde{X}_1 \cdot \tilde{Y}_1} \cdot \overline{\tilde{X}_2 \cdot \tilde{Y}_2} \mathbf{Cov}(\tilde{X}_3, \tilde{Y}_3) \right. \\ &\quad \left. + \overline{\tilde{X}_3 \cdot \tilde{Y}_3} \mathbf{Cov}(\tilde{X}_1, \tilde{Y}_1) \mathbf{Cov}(\tilde{X}_2, \tilde{Y}_2) + \overline{\tilde{X}_1 \cdot \tilde{Y}_1} \mathbf{Cov}(\tilde{X}_2, \tilde{Y}_2) \mathbf{Cov}(\tilde{X}_3, \tilde{Y}_3) \right. \\ &\quad \left. + \overline{\tilde{X}_2 \cdot \tilde{Y}_2} \mathbf{Cov}(\tilde{X}_1, \tilde{Y}_1) \mathbf{Cov}(\tilde{X}_3, \tilde{Y}_3) + \mathbf{Cov}(\tilde{X}_1, \tilde{Y}_1) \mathbf{Cov}(\tilde{X}_2, \tilde{Y}_2) \mathbf{Cov}(\tilde{X}_3, \tilde{Y}_3) \right\}. \tag{B.6} \end{aligned}$$

The terms appearing in the last expression are readily seen to be equal to

$$\begin{aligned}\overline{\tilde{X}_1} &= \overline{\tilde{X}_2} = \overline{\tilde{Y}_1} = \overline{\tilde{Y}_2} = 1 - 2p \\ \overline{\tilde{X}_3} &= \overline{\tilde{Y}_3} = 1 - 2q \\ \mathbf{Cov}(\tilde{X}_k, \tilde{Y}_k) &= 4\Sigma_r(\Gamma_{ij}^{(k)}, t_r, d) \text{ for } k \in \{1, 2\} \\ \mathbf{Cov}(\tilde{X}_3, \tilde{Y}_3) &= 4\alpha\end{aligned}$$

Plugging these expressions in (B.6) yields (4.5). $\square$

B.2 Proofs for §4.2

We prove in this subsection Theorem 4.7

Theorem 4.7 (DFR of the concatenated code). *The DFR of the concatenated code, using a $[n_e, k_e, d_e]_{\mathbb{F}_{256}}$ Reed-Solomon code as external code and an internal code of length n_i , for an error of weight t is :*

$$\text{DFR}(t) = \frac{1}{\binom{n}{t}} \sum_{\ell=\delta_e+1}^{n_e} a_{t,\ell}$$

with

$$\begin{aligned}a_{t,\ell} &= [X^t Y^\ell] \left(\left[\sum_{j=0}^{n_i} \binom{n_i}{j} (1 - \text{DFR}_i(j)) X^j + Y \cdot \sum_{j=0}^{n_i} \binom{n_i}{j} \text{DFR}_i(j) X^j \right]^{n_e} \right) \\ &= [X^t] \left(\binom{n_e}{\ell} \left[\sum_{j=0}^{n_i} \binom{n_i}{j} \text{DFR}_i(j) X^j \right]^\ell \left[\sum_{j=0}^{n_i} \binom{n_i}{j} (1 - \text{DFR}_i(j)) X^j \right]^{n_e - \ell} \right)\end{aligned}$$

where $n = n_e n_i$, $\delta_e = \lfloor \frac{d_e - 1}{2} \rfloor = \lfloor \frac{n_e - k_e}{2} \rfloor$ and, for $0 \leq j \leq n_i$, $\text{DFR}_i(j)$ is the DFR of the internal code for an error of weight j .

Proof. Let $\mathbf{e}$ be a random variable uniformly distributed over the set of binary vectors of length n and weight t . The positions in $\llbracket 1, n \rrbracket$ are divided in n_e blocks of n_i positions. For $1 \leq k \leq n_e$, let J_k be the weight of $\mathbf{e}$ restricted to the k^{th} block, and F_k be the event “the internal code fails to decode the k^{th} block”.

The concatenated code fails to decode if and only if the Reed-Solomon code has strictly more than δ_e errors left to decode, therefore

$$\text{DFR}(t) = \sum_{\ell=\delta_e+1}^{n_e} \binom{n_e}{\ell} \mathbf{P} \left(\bigcap_{k=1}^{\ell} F_k \cap \bigcap_{k=\ell+1}^{n_e} \overline{F_k} \right)$$

Now we have that :

$$\begin{aligned}& \mathbf{P} \left(\bigcap_{k=1}^{\ell} F_k \cap \bigcap_{k=\ell+1}^{n_e} \overline{F_k} \right) \\ &= \sum_{\substack{0 \leq j_1 \leq n_i \\ j_1 + \dots + j_{n_e} = t}} \mathbf{P}(J_1 = j_1, \dots, J_{n_e} = j_{n_e}) \mathbf{P} \left(\bigcap_{k=1}^{\ell} F_k \cap \bigcap_{k=\ell+1}^{n_e} \overline{F_k} \mid J_1 = j_1, \dots, J_{n_e} = j_{n_e} \right)\end{aligned}$$

with

$$\mathbf{P}(J_1 = j_1, \dots, J_{n_e} = j_{n_e}) = \frac{\prod_{k=1}^{n_e} \binom{n_i}{j_k}}{\binom{n}{t}}$$

and

$$\begin{aligned} \mathbf{P} \left(\bigcap_{k=1}^{\ell} F_k \cap \bigcap_{k=\ell+1}^{n_e} \overline{F_k} \mid J_1 = j_1, \dots, J_{n_e} = j_{n_e} \right) &= \prod_{k=1}^{\ell} \mathbf{P}(F_k \mid J_k = j_k) \cdot \prod_{k=\ell+1}^{n_e} \mathbf{P}(\overline{F_k} \mid J_k = j_k) \\ &= \prod_{k=1}^{\ell} \text{DFR}_i(j_k) \prod_{k=\ell+1}^{n_e} (1 - \text{DFR}_i(j_k)) \end{aligned}$$

since the internal decoding on one block is independent of the other blocks.

Thus :

$$\begin{aligned} \text{DFR}(t) &= \binom{n}{t} \sum_{\ell=\delta_e+1}^{n_e} \binom{n_e}{\ell} \sum_{\substack{0 \leq j \leq n_i \\ j_1 + \dots + j_{n_e} = t}} \left(\prod_{k=1}^{\ell} \binom{n_i}{j_k} \text{DFR}_i(j_k) \right) \cdot \left(\prod_{k=\ell+1}^{n_e} \binom{n_i}{j_k} (1 - \text{DFR}_i(j_k)) \right) \\ &= \binom{n}{t} \sum_{\ell=\delta_e+1}^{n_e} a_{t,\ell} \end{aligned}$$

□